Online appendix

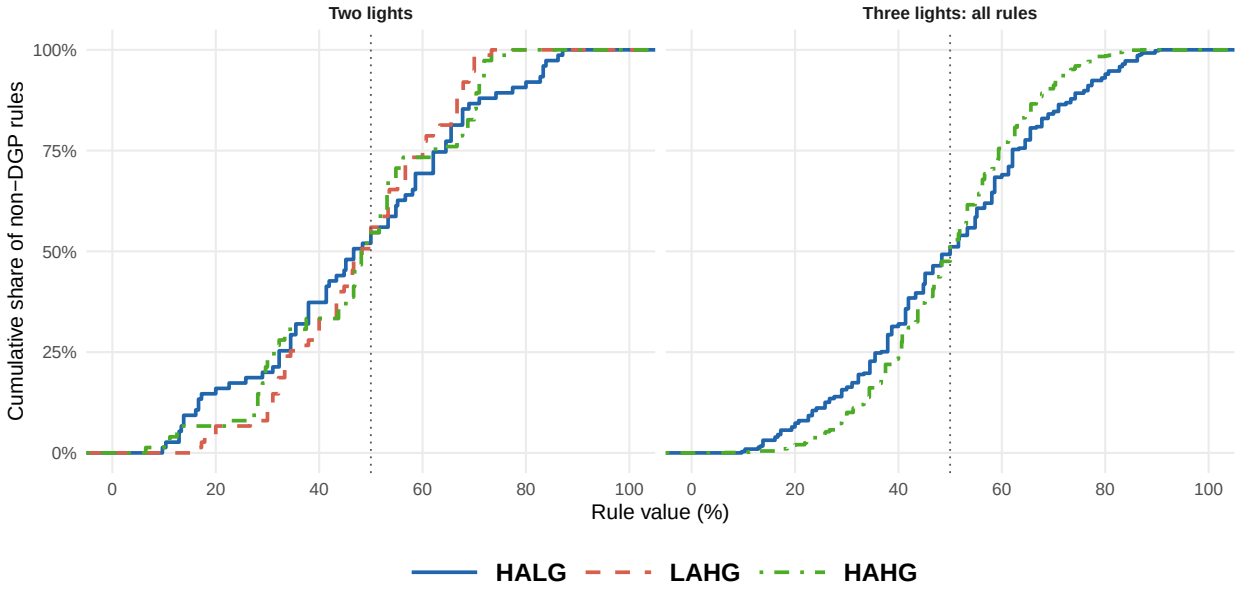
OA ADDITIONAL FIGURES AND TABLES

OA(i) DATA DESIGN

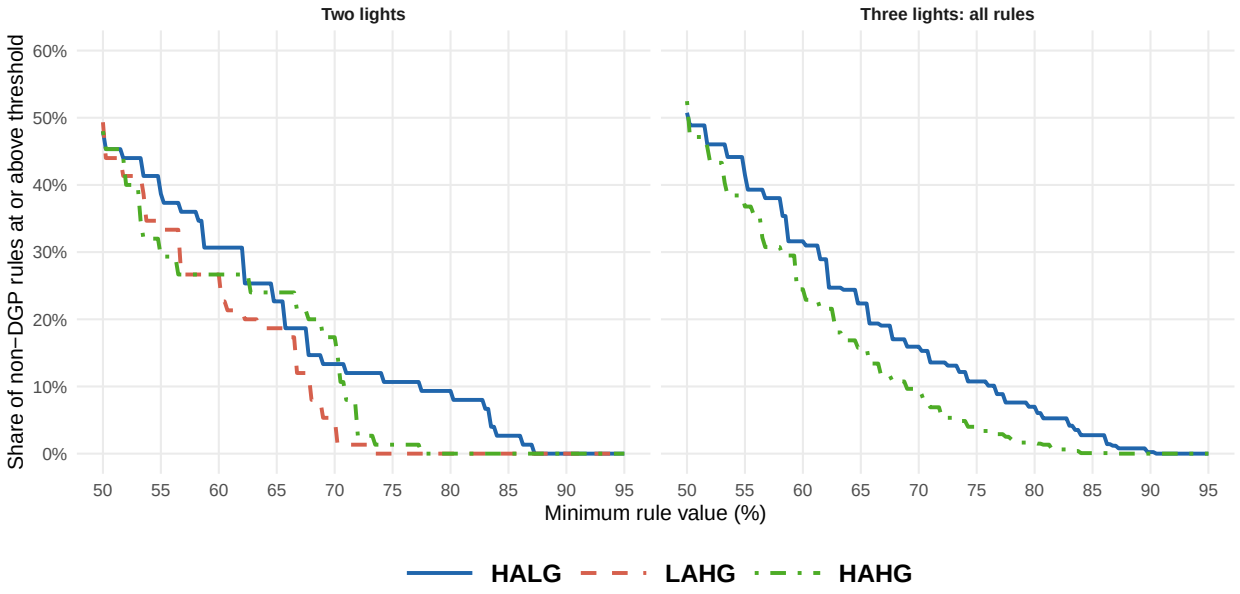
	Two lights			Three lights (all rules)	
	HALG	LAHG	HAHG	HALG	HAHG
Mean non-DGP rule value (%)	47.47	47.89	47.37	49.85	49.85
Standard deviation (p.p.)	21.21	14.86	17.91	18.87	14.35
10th percentile (%)	16.13	30.00	28.12	23.33	30.00
90th percentile (%)	77.42	67.86	70.97	76.67	68.75
90th–10th percentile spread (p.p.)	61.29	37.86	42.84	53.33	38.75

Notes: The unit of observation is a rule not associated to the DGP in one machine-specific dataset. The two-light columns pool the 15 non-DGP rules from each of five machines; the three-light columns pool the 255 non-DGP rules from each of five machines. Thus, every machine receives equal weight. Rule value is the rule's in-sample accuracy. The standard deviation measures dispersion in rule value. The 10th and 90th percentiles are the empirical lower- and upper-decile values, and the interdecile range is their difference, $P_{90} - P_{10}$. The DGP rule itself is excluded from every statistic.

Table OA1: Distribution of non-DGP rule values by accuracy-gap condition



(a) Empirical CDFs of alternative rule values



(b) Upper-tail survival functions

Figure OA1: Distribution of non-DGP rule values by accuracy-gap condition and predictor set. Panel (a) reports empirical CDFs of in-sample accuracy among non-DGP deterministic rules. Panel (b) reports upper-tail survival functions: at each threshold, the share of non-DGP rules with value at or above that threshold. The two-light panels pool the 15 non-DGP rules from each of five machines; the three-light panels pool the 255 non-DGP rules from each of five machines, giving each machine equal weight. Vertical dotted lines mark random-choice accuracy (50 percent).

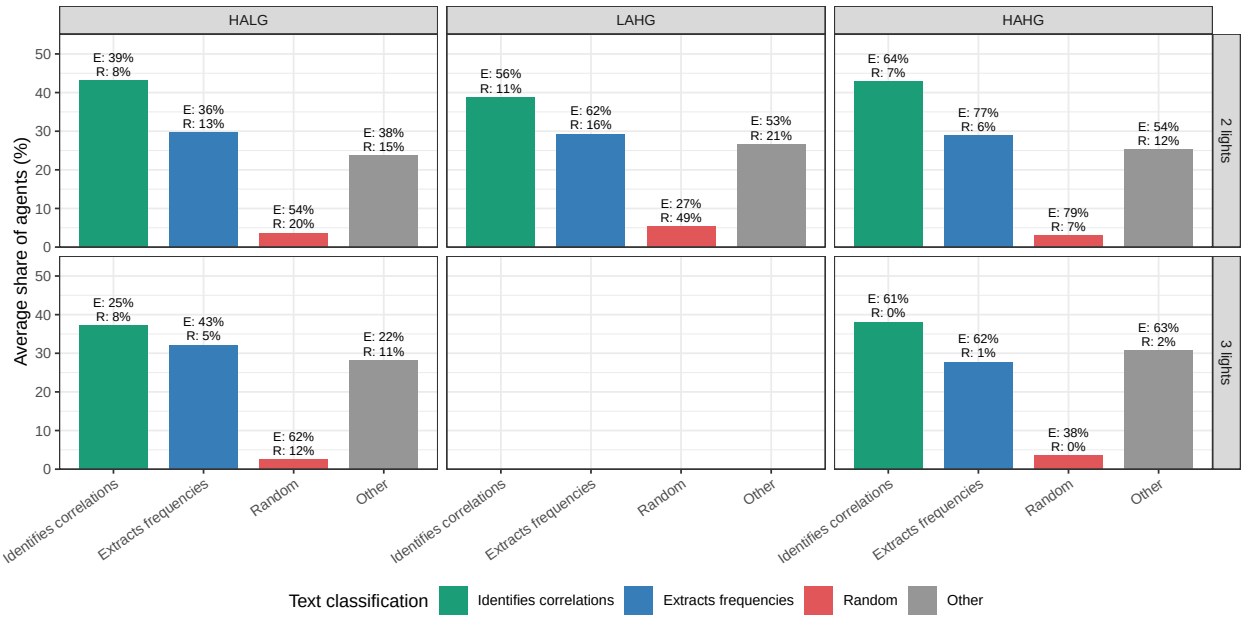


Figure OA2: Written-note classifications by experiment and accuracy-gap condition. Bars report the average shares of subject–dataset observations whose notes identify correlations, extract frequencies, describe random behavior, or fall into the residual category. Labels report, within each text category, the shares classified by the exact-tests as extracting the highest-value rule (E) or randomizing (R).

OB EX ANTE INFORMATIVENESS: PARALLELS WITH INFORMATION ECONOMICS

The objective of this section is to establish a systematic connection between a decision problem and a metric that captures the difference in value between the highest-value rule and other rules. Throughout, we study the misspecified-model case described in Online Appendix OC and assume $m(\cdot \mid x_M) = p(\cdot \mid x_M)$ for every x_M and M. We restrict attention to dataset–model pairs for which the conditional theory is uniquely identified and induces an optimal action.

We begin by relating the primitives of our framework to those of a canonical information design problem. An *experiment* is usually defined as a mapping $h : \Omega \to \Delta(\mathcal{S})$, where Ω denotes a finite state space and $\mathcal{S}$ a finite signal space. Note that this definition of experiments is prior-free. Ultimately, in a Bayesian framework, decision makers have priors $\mu_0 \in \text{int}(\Delta(\Omega))$ when they make decisions. In such cases an agent would then use Bayes' rule to update her beliefs: that is, for a signal s that occurs with positive probability given h and μ_0, one would have

$$\tau_h(s)\mu(\omega \mid s) = \mu_0(\omega)h(s \mid \omega)$$

where

$$\tau_h(s) = \sum_{\omega \in \Omega} \mu_0(\omega)h(s \mid \omega) \qquad \text{and} \qquad \mu(\omega \mid s) = \frac{h(s \mid \omega)\mu_0(\omega)}{\sum_{\omega' \in \Omega} h(s \mid \omega')\mu_0(\omega')}$$

In our framework, the outcome space Y can be interpreted as the state space Ω. Recall that a model M pins down a space of variables X_M, and that each realization x_M conveys information in the form of a theory $m(y \mid x_M)$. While a realization x_M conceptually corresponds to what the information economics literature typically refers to as a *signal realization*, and X_M to the corresponding *signal space*, some caution is needed in interpreting the direction of inference. In particular, in a standard information design problem an experiment specifies a probability distribution over signal realizations conditional on states which, together with a prior over the states and Bayes' rule, induces a posterior distribution. By contrast, in our framework the unconditional distribution over X is a primitive and, jointly with the DGP $m^*(y \mid \cdot)$, determines the unconditional frequency of y. In standard information design the prior distribution over Y is itself a primitive.

As noted by Cabrales et al. (2013), the ranking of experiments typically depends on two key considerations. First, the agent's prior beliefs over the state play a central role: for example, an agent who is nearly convinced that a severe crisis in the strength of the dollar is imminent may interpret adverse financial news quite differently from an agent who holds a more moderate view regarding the likelihood of such a crisis. Second, the agent's preferences (and wealth, when it comes to choices under risk) also matter. For instance, two

agents with identical priors but different risk preferences may rank differently a signal that nearly eliminates uncertainty about a rare financial loss versus one that refines beliefs about more probable events. Because decision-makers may hold different preferences, different decision problems might assign different values to the same experiment. Consequently, it is not possible to construct a complete ordering of information structures that applies universally. Specifically, expanding the set of priors or the set of utility functions leads to a more partial ordering on experiments, that is, it is less likely that any two signal distributions will be preferred by all decision-makers considered. This intuition translates directly to our framework.

In Section OB(i), we show that the DGP is more informative than other possible theories across all decision problems, drawing a parallel between the ranking of experiments in Blackwell (1953) and our framework. In Section OB(ii), we instead relate a notion of informativeness in instrumental decision-making to a specific class of decision problems.

OB(i) ALL UTILITIES AND PRIORS

The foundational result provided by Blackwell (1953) for the standard notion of experiments also applies within our framework. According to Blackwell's criterion, one experiment is more informative than another if and only if every decision maker, regardless of utility function, would prefer the more informative experiment in any decision problem. This provides a decision-theoretic foundation for the informativeness ordering. Formally, for a model M, define the state-to-signal kernels

$$q_M(x_M \mid y) := p(x_M \mid y)$$

for states y of positive prior probability. Then

$$q_M(x_M \mid y) = \sum_{x_{M^*}} p(x_M \mid x_{M^*}) q_{M^*}(x_{M^*} \mid y).$$

Thus q_M is a garbling of q_{M^*}, and X_{M^*} has weakly greater Bayes value than X_M in every finite decision problem. Therefore, for any prior and utility function, a decision maker obtains weakly greater Bayes value from the experiment q_{M^*} than from q_M, when choosing an optimal action conditional on the realized signal. Although the experiments associated with two misspecified models may be ranked according to the Blackwell order in special cases, such a ranking does not generally exist and may depend on the dataset. By contrast, for every dataset $p \in \mathcal{P}_{m^*}$, the experiment q_{M^*} Blackwell-dominates q_M for every model M, which justifies our focus on agents learning the true model M^*. In our framework, this Blackwell dominance translates into a weakly higher value for the highest-value rule

associated with the true model: for every dataset $p \in \mathcal{P}_{m^*}$ and every model $M \in \mathcal{P}(N)$,

$$v^{\mathcal{D}}(\pi^{\max}_{M^*}) \geq v^{\mathcal{D}}(\pi^{\max}_{M}) \,.$$

OB(ii) THE VALUE OF INFORMATION

One important notion in our framework is given by the difference in value between the highest-value rule and another rule that maximizes value conditional on its model, $v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\pi^{\max}_{M})$. In this section, we formalize the notion of difference and relate it to what is known in the literature as the *value of information*, showing that, in the case of misspecified models, this difference can be easily computed.

The value of information. Define the value of adopting the best action according to $m(y \mid x_M)$ when the DGP is $m^*(y \mid x_{M^*})$ as

$$V^{\mathcal{D}}_{m^*,m}(x_M, x_{M^*}) = \sum_{y \in Y} m^*(y \mid x_{M^*}) \cdot u\Big(\hat{a}(x_M), y\Big).$$

The *value of information* is defined as

$$L^{\mathcal{D}}_{m^*,m}(x_M, x_{M^*}) = v^{\mathcal{D}}(\hat{a}_{M^*}(x_{M^*})) - V^{\mathcal{D}}_{m^*,m}(x_M, x_{M^*}),$$

where $\hat{a}_{M^*}(x_{M^*})$ denotes the optimal action associated with the true model. This definition is the same as the value of information introduced in the literature on information design (Gossner and Steiner, 2018, Frankel and Kamenica, 2019). We are interested in an ex ante notion of *value of information*, defined as:

$$L^{\mathcal{D}}(\pi^{\max}_{M^*}, \pi^{\max}_{M}) = \sum_{x_M} \left(\sum_{x_{M^*}} p(x_{M^*} \mid x_M) L^{\mathcal{D}}_{m^*,m}(x_M, x_{M^*}) \right) p(x_M).$$

Lemma 1. *The ex ante value of information can be written as* $L^{\mathcal{D}}(\pi^{\max}_{M^*}, \pi^{\max}_{M}) = v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\pi^{\max}_{M})$.

Proof. The ex ante value of information is defined as:

$$\begin{aligned} L^{\mathcal{D}}(\pi^{\max}_{M^*}, \pi^{\max}_{M}) &= v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \mathbb{E}_p \left[\sum_{y \in Y} m^*(y \mid x_{M^*}) \cdot u(\hat{a}_M(x_M), y) \right] \\ &= v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\pi^{\max}_{M}) \end{aligned}$$

where the second equality comes from the law of iterated expectations. □

Corollary 3. *For every model M,*

$$v^{\mathcal{D}}(\pi_M^{\max}) - v^{\mathcal{D}}(\pi_\emptyset^{\max}) = L^{\mathcal{D}}\left(\pi_{M^*}^{\max}, \pi_\emptyset^{\max}\right) - L^{\mathcal{D}}\left(\pi_{M^*}^{\max}, \pi_M^{\max}\right).$$

Lemma 1 connects the standard ex ante notion of the value of information to a natural analogue in our framework. This result is driven by the assumption that the agent adopts an unbiased estimator for every theory, $m(y \mid x_M) = p(y \mid x_M)$, which can be interpreted as a Bayesian consistency condition.

OB(iii) SIMPLE DECISION PROBLEMS

In what follows we consider simple decision problems with a binary outcome space, since the utility in such problems resembles the way subjects are incentivized in the experiment. We maintain that agents adopt the correct theory, that is, $m(\cdot \mid x_M) = p(\cdot \mid x_M)$ for all $x_M \in X_M$ and all $M \in \mathcal{P}(N)$.

The outcome space is binary, with $Y = \{0, 1\}$, and the action space is also binary, with $A = \{a_0, a_1\}$. We assume that a_0 is optimal when the outcome is $y = 0$, so that $u(a_0, 0) > u(a_1, 0)$, and that a_1 is optimal when the outcome is $y = 1$, with $u(a_1, 1) > u(a_0, 1)$. One example is the case in which $A = \{0, 1\}$ and the agent's objective is to match the realized outcome of the world $Y = \{0, 1\}$.[59] Another interpretation is that A represents actions, such as investing or not investing, that are optimal in different states.

Denote by m and $m(x_M)$ the belief and the conditional belief (respectively) that the outcome is $y = 1$, that is, $m(1 \mid x_M)$. Normalizing the payoff of a_0 to zero in both states, let $\Delta u \equiv u(a_1, 1) - u(a_1, 0) > 0$. Then a_1 is an optimal action if $m \cdot u(a_1, 1) + (1-m) \cdot u(a_1, 0) \geq 0$, i.e., $m \geq \bar{m} \equiv -u(a_1, 0)/\Delta u$. Upon adopting model M and observing the realization x_M, the agent forms belief $m(x_M)$ and achieves a payoff of $\Delta u \left(m(x_M) - \bar{m}\right)$ if the optimal action is a_1 and 0 otherwise. The expected payoff when the agent adopts model M, that is the value of rule $\pi_M^{\max}$ is given by

$$v^{\mathcal{D}}(\pi_M^{\max}) = \Delta u \sum_{x_M} p(x_M) \left(m(x_M) - \bar{m}\right) \cdot \mathbf{1}_{m(x_M) > \bar{m}}$$

Conditional on the realizations x_M and x_{M^*}, the value of using the information contained in M^* rather than M depends on whether the two models prescribe different actions. Specifically, the gain is:

(i) $\Delta u \left(m^*(x_{M^*}) - \bar{m}\right)$ if $m(x_M) < \bar{m} \leq m^*(x_{M^*})$, in which case the prescribed action changes from a_0 to a_1;

[59] In such experiments, utility is given by $u(a, y) = B \cdot \mathbf{1}_{\{a=y\}}$, where B denotes the monetary incentive.

(ii) $\Delta u\,(\bar{m} - m^*(x_{M^*}))$ if $m^*(x_{M^*}) < \bar{m} \leq m(x_M)$, in which case the prescribed action changes from a_1 to a_0;

(iii) 0 if

$$\max\{m(x_M), m^*(x_{M^*})\} < \bar{m} \quad \text{or} \quad \min\{m(x_M), m^*(x_{M^*})\} \geq \bar{m},$$

in which case the two models prescribe the same action.

On the ordered-pair domain $X_M \times X_{M^*}$, define the two action-switching events by

$$E_+^{\mathcal{D}^{\bar{m}}}(M, M^*) \equiv \{(x_M, x_{M^*}) \in X_M \times X_{M^*} : m(x_M) < \bar{m} \leq m^*(x_{M^*})\},$$
$$E_-^{\mathcal{D}^{\bar{m}}}(M, M^*) \equiv \{(x_M, x_{M^*}) \in X_M \times X_{M^*} : m^*(x_{M^*}) < \bar{m} \leq m(x_M)\}.$$

The ex ante value of information is therefore

$$L^{\mathcal{D}^{\bar{m}}}(\pi^{\max}_{M^*}, \pi^{\max}_{M}) = \Delta u \sum_{(x_M, x_{M^*}) \in X_M \times X_{M^*}} p(x_M, x_{M^*}) \times \left[(m^*(x_{M^*}) - \bar{m})\, \mathbf{1}_{E_+^{\mathcal{D}^{\bar{m}}}(M, M^*)} + (\bar{m} - m^*(x_{M^*}))\, \mathbf{1}_{E_-^{\mathcal{D}^{\bar{m}}}(M, M^*)} \right].$$

The effect can be decomposed into two parts, with the first depending on the magnitude of the error when it occurs, and the second given by the probability of committing the error, as evident in the following equation:

$$L^{\mathcal{D}^{\bar{m}}}(\pi^{\max}_{M^*}, \pi^{\max}_{M}) = \Delta u \Bigg[\left(\mathbb{E}[m^*(x_{M^*}) \mid E_+^{\mathcal{D}^{\bar{m}}}(M, M^*)] - \bar{m} \right) \cdot \mathbb{P}\left(E_+^{\mathcal{D}^{\bar{m}}}(M, M^*) \right) + \left(\bar{m} - \mathbb{E}[m^*(x_{M^*}) \mid E_-^{\mathcal{D}^{\bar{m}}}(M, M^*)] \right) \cdot \mathbb{P}\left(E_-^{\mathcal{D}^{\bar{m}}}(M, M^*) \right) \Bigg].$$

For a fixed probability of choosing the wrong action, a greater distance between the true conditional belief $m^*(x_{M^*})$ and the decision threshold $\bar{m}$ results in a larger expected loss; likewise, holding the loss conditional on an error fixed, a higher probability of choosing the wrong action results in a larger expected loss.

OC DIFFERENT TYPES OF MISSPECIFICATION

Here we formalize the idea of agents choosing a rule by relying on (i) a model M and (ii) a *subjective theory*, defined as

$$m : X_M \to \Delta(Y),$$

which depends only on realizations of the variables $X_M = \prod_{i \in M} X_i$ indexed by a model $M \in \mathcal{P}(N)$. Given these beliefs, the agent chooses an action $a \in A$ to maximize value, with

$$\hat{a}(x_M) \in \arg\max_{a \in A} \sum_{y \in Y} u(a, y) \cdot m(y \mid x_M).$$

Define the expected value attained by this choice when x_M is realized as

$$v^{\mathcal{D}}(\hat{a}(x_M)) = \sum_{y \in Y} u(\hat{a}(x_M), y) \cdot p(y \mid x_M),$$

with $p(y \mid x_M)$ restricted to positive probability conditioning events.

Since the agent maximizes expected utility under m for every $x_M \in X_M$, we denote the induced rule by $\pi^{\max}_{(m,M)} : X_M \to \Delta(A)$, which allows us to define the expected value of adopting model M and subjective theory m in decision problem $\mathcal{D}$ as

$$v^{\mathcal{D}}(\pi^{\max}_{(m,M)}) = \sum_{x_M \in X_M} v^{\mathcal{D}}(\hat{a}(x_M)) \cdot p(x_M).$$

In what follows, we highlight two forms of misspecification that can arise in this framework. Specifically, the agent may be misspecified in the choice of the model, with $M \neq M^*$, in the choice of the subjective theory, $m(y \mid \cdot) \neq p(y \mid \cdot)$, or in both. The distinction between misspecification in M while using the correct conditional theory and misspecification in the theory itself parallels the contrast between the "misspecified model" approach and the "non-Bayesian" approach in the literature on information economics (Bohren and Hauser, 2023). The misspecified model approach offers a general methodology for capturing behavioral biases: without departing from the standard rational-expectations paradigm, it allows agents to hold a subjective, potentially incorrect, model of the signal-generating process, while maintaining the assumption that beliefs are updated according to Bayes' rule (see Bohren and Hauser (2025) for a recent review). By contrast, the non-Bayesian framework directly parameterizes the updating rule, mapping signal realizations into posterior beliefs without reference to an underlying model (e.g., Woodford, 2020; Ba et al., 2026). While the non-Bayesian framework provides a transparent connection between the form of the error and the resulting distortions in beliefs, it is typically applied in a case-specific manner and therefore lacks the generality of the misspecified model approach. The next paragraphs formalize this distinction in our framework.

Misspecified Models. We first consider the case in which the agent can be misspecified in the space of models, but correctly uses the data to infer the theory. If all possible misspecified theories are permitted, the framework loses empirical content: its empirical validity hinges on the identification of the agent's subjective theory and on relating the perceived value of

the associated rule to the primitives of the environment. Our objective is to relate the choice of a model M and the associated theory by imposing a consistency requirement with the data. To this end, we restrict attention to model misspecification, assuming that the agent may use $M \neq M^*$ but, conditional on this choice, forms the subjective theory consistently with the observed dataset $p \in \mathcal{P}_{m^*}$ and chooses the optimal action. That is, we assume $m(\cdot \mid x_M) = p(\cdot \mid x_M)$ for all $x_M \in X_M$ and all $M \in \mathcal{P}(N)$. Under this framework, we denote the value-maximizing rule conditional on model M by $\pi_M^{\max}$. Online Appendix OB derives several results linking ex ante measures of informativeness to this form of misspecification, showing that this restriction on how the agent may be misspecified provides a rigorous and disciplined framework for analyzing misspecification.

Misspecified Theories. An alternative approach is to consider an agent who is also misspecified in the space of theories. In this paragraph, we outline two relevant cases. First, as emphasized by the literature on imperfect encoding and cognitive noise, the agent's assessments could be inaccurate.[60] We can model this by allowing for mistakes in assessing the conditional probability of the outcome. Consider a binary outcome space $Y = \{0, 1\}$. Using the noisy-encoding specification in Khaw et al. (2022), we can define the log odds

$$z = \log\left(\frac{p(1 \mid x_M)}{p(0 \mid x_M)}\right),$$

with $p(1 \mid x_M) \in (0, 1)$, and denote the encoded log odds as

$$r_p = z + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2). \tag{14}$$

We then define $m(1 \mid x_M) = \frac{\exp(r_p)}{1+\exp(r_p)}$.

We note, however, that in models of cognitive imprecision stochasticity originates entirely from the noisy encoding, whereas our framework attributes it to the costly extraction of rules from structured data. The two approaches thus connect only through the perception of value, while the comparative statics differ: cognitive imprecision holds the environment fixed and ties mistakes to perceptual noise, whereas our framework turns on the value gap between the highest-value rule and its competitors, as determined by the environment's correlational structure.

Another important source of misperception arises when an agent adopts a deterministic theory for model M while the DGP is stochastic, a phenomenon known as rounding (Ruud et al., 2014). In this case, the subjective theory is given by

$$m(1 \mid x_M) = \mathbb{1}\{\, p(1 \mid x_M) \;\geq\; \bar{m} \,\}, \tag{15}$$

[60] See Woodford (2020) and Enke (2024) for a review.

with $\bar{m}$ being an arbitrary threshold. Future research may further link rules to the ways in which boundedly rational agents face limitations not only in model choice, but also in the selection of the theory m associated with a given model. While we view this avenue as more "applied" and case-specific in nature, it can be used to shed light on many problems.

OD LEMMAS AND ADDITIONAL PROPOSITIONS

OD(i) Lemmas

Lemma 2. *Suppose that $\mathcal{D}$ and $\mathcal{D}'$ have the same candidate rule set and cost function. If $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') \geq 1$ for every $\pi_M \in \Pi$, then $\Lambda(\Pi^{\mathcal{D}'}) \geq \Lambda(\Pi^{\mathcal{D}})$. The inequality is strict if $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') > 1$ for some $\pi_M \in \Pi^{\mathcal{D}}$. The reverse inequalities hold when $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') \leq 1$ for every $\pi_M \in \Pi$, with strict inequality if $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') < 1$ for some $\pi_M \in \Pi^{\mathcal{D}'}$.*

Proof. First, suppose that $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') \geq 1$ for every $\pi_M \in \Pi$. Since rule values are strictly positive, this implies $v^{\mathcal{D}'}(\pi_M) \geq v^{\mathcal{D}}(\pi_M)$ for every $\pi_M \in \Pi$.

Suppose, toward a contradiction, that $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$. Then $\Pi^{\mathcal{D}} \subseteq \Pi^{\mathcal{D}'}$. Indeed, if $\pi_M \in \Pi^{\mathcal{D}}$, then

$$v^{\mathcal{D}'}(\pi_M) \geq v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}}) > \Lambda(\Pi^{\mathcal{D}'}),$$

and hence $\pi_M \in \Pi^{\mathcal{D}'}$.

Using equation (4) and the fact that extraction probabilities sum to one, we obtain

$$\begin{aligned} 1 &= \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi_M)} \geq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi_M)} \\ &> \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} = 1. \end{aligned}$$

The strict inequality follows from $v^{\mathcal{D}'}(\pi_M) \geq v^{\mathcal{D}}(\pi_M)$ and $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$ for every $\pi_M \in \Pi^{\mathcal{D}}$. This is a contradiction. Therefore, $\Lambda(\Pi^{\mathcal{D}'}) \geq \Lambda(\Pi^{\mathcal{D}})$.

We next establish the condition for strict inequality. Suppose that $\Delta^{\widehat{\pi}_{\widehat{M}}}(\mathcal{D}, \mathcal{D}') > 1$ for some $\widehat{\pi}_{\widehat{M}} \in \Pi^{\mathcal{D}}$ and, toward a contradiction, that $\Lambda(\Pi^{\mathcal{D}'}) \leq \Lambda(\Pi^{\mathcal{D}})$. As above, this implies $\Pi^{\mathcal{D}} \subseteq \Pi^{\mathcal{D}'}$. Consequently,

$$\begin{aligned} 1 &= \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi_M)} \geq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi_M)} \\ &\geq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} > \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} = 1. \end{aligned}$$

The final inequality is strict because $v^{\mathcal{D}'}(\widehat{\pi}_{\widehat{M}}) > v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}})$ and $\widehat{\pi}_{\widehat{M}} \in \Pi^{\mathcal{D}}$. This contradiction proves that $\Lambda(\Pi^{\mathcal{D}'}) > \Lambda(\Pi^{\mathcal{D}})$.

Finally, suppose that

$$\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') \leq 1 \qquad \text{for every } \pi_M \in \Pi.$$

Then

$$v^{\mathcal{D}}(\pi_M) \geq v^{\mathcal{D}'}(\pi_M) \qquad \text{for every } \pi_M \in \Pi.$$

Applying the preceding argument with $\mathcal{D}$ and $\mathcal{D}'$ interchanged gives $\Lambda(\Pi^{\mathcal{D}'}) \leq \Lambda(\Pi^{\mathcal{D}})$. Moreover, if $\Delta^{\widehat{\pi}_{\widehat{M}}}(\mathcal{D}, \mathcal{D}') < 1$ for some $\widehat{\pi}_{\widehat{M}} \in \Pi^{\mathcal{D}'}$, then $v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) > v^{\mathcal{D}'}(\widehat{\pi}_{\widehat{M}})$, and the strict part of the preceding argument, again with the two decision problems interchanged, yields $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$. □

Now denote by $\Pi^{\mathcal{D}}_{\kappa}$ the consideration set when the cost function is $\kappa(\cdot)$.

Lemma 3. *If the cost function is scaled by a factor $\gamma > 1$ (i) for all $\pi_M \in \Pi$ or (ii) only for the highest-value rule $\pi^{\max}_{M^*}$ then $\Lambda(\Pi^{\mathcal{D}}_{\kappa}) > \Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa})$*

Proof. For part (i), suppose, toward a contradiction, that

$$\Lambda\left(\Pi^{\mathcal{D}}_{\gamma\kappa}\right) \geq \Lambda\left(\Pi^{\mathcal{D}}_{\kappa}\right).$$

Since rule values are unchanged, this implies $\Pi^{\mathcal{D}}_{\gamma\kappa} \subseteq \Pi^{\mathcal{D}}_{\kappa}$. Indeed, if $\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa}$, then

$$v^{\mathcal{D}}(\pi_M) > \Lambda\left(\Pi^{\mathcal{D}}_{\gamma\kappa}\right) \geq \Lambda\left(\Pi^{\mathcal{D}}_{\kappa}\right),$$

and hence $\pi_M \in \Pi^{\mathcal{D}}_{\kappa}$. Using the fact that extraction probabilities sum to one, we obtain

$$\begin{aligned}
1 &= \sum_{\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda\left(\Pi^{\mathcal{D}}_{\gamma\kappa}\right)}{\gamma\kappa(\pi_M)} \\
&< \sum_{\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda\left(\Pi^{\mathcal{D}}_{\kappa}\right)}{\kappa(\pi_M)} \\
&\leq \sum_{\pi_M \in \Pi^{\mathcal{D}}_{\kappa}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda\left(\Pi^{\mathcal{D}}_{\kappa}\right)}{\kappa(\pi_M)} = 1,
\end{aligned}$$

which is a contradiction. The strict inequality follows because $\gamma > 1$ and, for every $\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa}$, $v^{\mathcal{D}}(\pi_M) - \Lambda\left(\Pi^{\mathcal{D}}_{\gamma\kappa}\right) > 0$. The weak inequality follows from $\Pi^{\mathcal{D}}_{\gamma\kappa} \subseteq \Pi^{\mathcal{D}}_{\kappa}$. Therefore,

$$\Lambda\left(\Pi^{\mathcal{D}}_{\gamma\kappa}\right) < \Lambda\left(\Pi^{\mathcal{D}}_{\kappa}\right).$$

For part (ii), suppose, toward a contradiction, that $\Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa}) \geq \Lambda(\Pi^{\mathcal{D}}_{\kappa})$, where $\Pi^{\mathcal{D}}_{\gamma\kappa}$ denotes the consideration set when only the cost of $\pi^{\max}_{M^*}$ is multiplied by γ. Since rule values are unchanged, this implies $\Pi^{\mathcal{D}}_{\gamma\kappa} \subseteq \Pi^{\mathcal{D}}_{\kappa}$. Moreover, as established above, the value-maximizing

rule $\pi^{\max}_{M^*}$ belongs to both consideration sets. Hence,

$$\begin{aligned}
1 &= \frac{v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa})}{\gamma\kappa(\pi^{\max}_{M^*})} + \sum_{\substack{\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa} \\ \pi_M \neq \pi^{\max}_{M^*}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa})}{\kappa(\pi_M)} \\
&< \frac{v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \Lambda(\Pi^{\mathcal{D}}_{\kappa})}{\kappa(\pi^{\max}_{M^*})} + \sum_{\substack{\pi_M \in \Pi^{\mathcal{D}}_{\gamma\kappa} \\ \pi_M \neq \pi^{\max}_{M^*}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}}_{\kappa})}{\kappa(\pi_M)} \\
&\leq \sum_{\pi_M \in \Pi^{\mathcal{D}}_{\kappa}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}}_{\kappa})}{\kappa(\pi_M)} = 1,
\end{aligned}$$

a contradiction. The strict inequality follows from $\gamma > 1$, $\Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa}) \geq \Lambda(\Pi^{\mathcal{D}}_{\kappa})$, and $v^{\mathcal{D}}(\pi^{\max}_{M^*}) > \Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa})$, whereas the weak inequality follows from $\Pi^{\mathcal{D}}_{\gamma\kappa} \subseteq \Pi^{\mathcal{D}}_{\kappa}$. Therefore, $\Lambda(\Pi^{\mathcal{D}}_{\gamma\kappa}) < \Lambda(\Pi^{\mathcal{D}}_{\kappa})$. □

OD(ii) Additional Propositions

We now turn to predictions that hinge directly on the assumption that the cost function reflects an *attention* allocation mechanism that is rule-specific, with simpler rules being easier to notice.

Proposition 5. *Consider two rules π_M and $\pi'_{M'}$ such that $\rho^{\mathcal{D}}(\pi_M) \in (0,1)$ and $\rho^{\mathcal{D}}(\pi'_{M'}) \in (0,1)$. Fix the decision problem $\mathcal{D}$, and let $\mathcal{D}'$ differ from $\mathcal{D}$ only in that $v^{\mathcal{D}'}(\pi_M) > v^{\mathcal{D}}(\pi_M)$. It then follows that:*

Direct effect: $\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M) > 0$ and, for $\Pi^{\mathcal{D}} = \Pi^{\mathcal{D}'}$, the difference is weakly increasing in,

(i) the value $v^{\mathcal{D}'}(\pi_M)$

(ii) the inverse cost $\frac{1}{\kappa(\pi_M)}$, for a fixed $v^{\mathcal{D}'}(\pi_M)$.

The difference is strictly increasing only while at least one competitor remains active.

Substitution patterns: $\rho^{\mathcal{D}}(\pi'_{M'}) - \rho^{\mathcal{D}'}(\pi'_{M'}) > 0$ and, for $\Pi^{\mathcal{D}} = \Pi^{\mathcal{D}'}$, the difference is greater,

(iii) The higher $v^{\mathcal{D}'}(\pi_M)$

(iv) For a fixed $v^{\mathcal{D}'}(\pi_M)$, the higher $\frac{1}{\kappa(\pi'_{M'})}$ and $\frac{1}{\kappa(\pi_M)}$

Point (ii) shows that changes in the value of a rule have a larger impact when its cost is low.[61] Point (iv) further establishes that changes in the value of a low-cost rule exert a

[61] Related insights appear in Krajbich et al. (2010), who study comparison processes in value-based decision making, and in Smith and Krajbich (2019), who characterize the interaction between value and saliency in the choice process and show that increases in saliency have stronger effects for high-utility items.

stronger effect on the choice probabilities of other rules, and that a rule with lower cost is itself more responsive to shifts in the values of competing rules. As a result, the probability of extracting the highest-value rule increases when it is simpler or when competing rules that change in value are simpler, since reductions in their values have a larger impact. Finally, when the values of all rules change simultaneously, due to substitution patterns, identification of the cost function becomes essential to recover the total effect, which is linear when the consideration set remains fixed.

Proof. First, we notice that increasing the value of rule π_M retains it in the consideration set for problem $\mathcal{D}'$. The difference in the probability of extracting the rule $\pi_M \in \Pi$ in problem $\mathcal{D}$ and $\mathcal{D}'$ can be therefore written as:

$$\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M) = \frac{1}{\kappa(\pi_M)} \left((\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') - 1) \cdot v^{\mathcal{D}}(\pi_M) - (\Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})) \right). \tag{16}$$

Point (i). Since only $v^{\mathcal{D}}(\pi_M)$ increases and $\pi_M \in \Pi^{\mathcal{D}}$, Lemma 2 implies $\Lambda(\Pi^{\mathcal{D}'}) > \Lambda(\Pi^{\mathcal{D}})$.

Every unchanged rule that is inactive under $\mathcal{D}$ remains inactive under $\mathcal{D}'$, because

$$v^{\mathcal{D}'}(\hat{\pi}) = v^{\mathcal{D}}(\hat{\pi}) \leq \Lambda(\Pi^{\mathcal{D}}) < \Lambda(\Pi^{\mathcal{D}'}).$$

Moreover, for every $\hat{\pi} \in \Pi^{\mathcal{D}} \setminus \{\pi_M\}$, either $\hat{\pi} \in \Pi^{\mathcal{D}'}$, in which case

$$\rho^{\mathcal{D}'}(\hat{\pi}) - \rho^{\mathcal{D}}(\hat{\pi}) = -\frac{\Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\hat{\pi})} < 0,$$

or $\hat{\pi} \notin \Pi^{\mathcal{D}'}$, in which case $\rho^{\mathcal{D}'}(\hat{\pi}) - \rho^{\mathcal{D}}(\hat{\pi}) = -\rho^{\mathcal{D}}(\hat{\pi}) < 0$.

It remains to verify that $\pi_M \in \Pi^{\mathcal{D}'}$. Suppose, toward a contradiction, that $\pi_M \notin \Pi^{\mathcal{D}'}$. Every rule active under $\mathcal{D}'$ would then be an unchanged competitor and would receive strictly less probability than under $\mathcal{D}$. Therefore,

$$1 = \sum_{\hat{\pi} \in \Pi^{\mathcal{D}'}} \rho^{\mathcal{D}'}(\hat{\pi}) < \sum_{\hat{\pi} \in \Pi^{\mathcal{D}'}} \rho^{\mathcal{D}}(\hat{\pi}) \leq 1 - \rho^{\mathcal{D}}(\pi_M) < 1,$$

a contradiction. Hence $\pi_M \in \Pi^{\mathcal{D}'}$.

Since $\rho^{\mathcal{D}}(\pi'_{M'}) > 0$, at least one competitor loses probability strictly. The simplex constraint therefore implies $\rho^{\mathcal{D}'}(\pi_M) > \rho^{\mathcal{D}}(\pi_M)$.

Point (ii)-(iv). If the consideration set does not change, that is, if $\Pi^{\mathcal{D}} = \Pi^{\mathcal{D}'}$, the marginal effects are linear and equal to:

$$\frac{\partial \rho^{\mathcal{D}}(\pi_M)}{\partial v(\pi_M)} = \frac{\frac{1}{\kappa(\pi_M)} \sum_{\pi'_{M'} \in \Pi^{\mathcal{D}} \setminus \{\pi_M\}} \frac{1}{\kappa(\pi'_{M'})}}{\sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi_M)}} > 0 \qquad \frac{\partial \rho^{\mathcal{D}}(\pi_M)}{\partial v(\pi'_{M'})} = \frac{\partial \rho^{\mathcal{D}}(\pi'_{M'})}{\partial v(\pi_M)} = -\frac{\frac{1}{\kappa(\pi_M)\kappa(\pi'_{M'})}}{\sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi_M)}} < 0.$$

□

We now turn to propositions about the cost. First, fixing a decision problem $\mathcal{D}$, we consider two rules that differ only in cost.

Proposition 6. *Consider two rules $\pi_M, \pi'_{M'} \in \Pi$ such that $v^{\mathcal{D}}(\pi_M) = v^{\mathcal{D}}(\pi'_{M'})$ and $\kappa(\pi_M) > \kappa(\pi'_{M'})$. Then $\rho^{\mathcal{D}}(\pi'_{M'}) - \rho^{\mathcal{D}}(\pi_M) \geq 0$, with strict inequality if $\rho^{\mathcal{D}}(\pi'_{M'}) > 0$.*

Holding all other rule values and all costs fixed, the difference is weakly increasing in the common value of the two rules. This monotonicity is strict whenever $\rho^{\mathcal{D}}(\pi'_{M'}) > 0$ and $\rho^{\mathcal{D}}(\pi_M) + \rho^{\mathcal{D}}(\pi'_{M'}) < 1$. For changes in either cost that leave the consideration set unchanged, the difference is weakly increasing in $\frac{1}{\kappa(\pi'_{M'})}$ and weakly decreasing in $\frac{1}{\kappa(\pi_M)}$. Both monotonicities are strict whenever $\rho^{\mathcal{D}}(\pi'_{M'}) > 0$.

Proposition 6 characterizes the difference in the probabilities of extracting two equally valuable rules with different costs within a fixed decision problem. The lower-cost rule is more likely to be extracted whenever either rule is extracted. Moreover, when at least one additional rule is active, this simplicity advantage increases with the common value attained by the two rules, reflecting a complementarity between value and cost.

Proof. Since the two rules attain the same value, either both belong to the consideration set or neither does. If neither rule belongs to $\Pi^{\mathcal{D}}$, both extraction probabilities are zero and the first claim follows.

Suppose instead that the two rules belong to $\Pi^{\mathcal{D}}$. Using equation (4) and $v^{\mathcal{D}}(\pi_M) = v^{\mathcal{D}}(\pi'_{M'})$, we obtain

$$\rho^{\mathcal{D}}(\pi'_{M'}) - \rho^{\mathcal{D}}(\pi_M) = \left(v^{\mathcal{D}}(\pi'_{M'}) - \Lambda(\Pi^{\mathcal{D}})\right)\left(\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}\right).$$

Both terms on the right-hand side are strictly positive. The first is positive because the rules belong to the consideration set, and the second is positive because $\kappa(\pi_M) > \kappa(\pi'_{M'})$. Hence $\rho^{\mathcal{D}}(\pi'_{M'}) > \rho^{\mathcal{D}}(\pi_M)$.

We next vary the common value of the two rules, holding all other values and all costs fixed. On any region over which the consideration set is unchanged and contains both rules,

$$\frac{d\Lambda(\Pi^{\mathcal{D}})}{dv^{\mathcal{D}}(\pi_M)} = \frac{\frac{1}{\kappa(\pi_M)} + \frac{1}{\kappa(\pi'_{M'})}}{\sum_{\hat{\pi}\in\Pi^{\mathcal{D}}} \frac{1}{\kappa(\hat{\pi})}},$$

where the two equal values are varied jointly. It follows that

$$\frac{d}{dv^{\mathcal{D}}(\pi_M)}\left[\rho^{\mathcal{D}}(\pi'_{M'}) - \rho^{\mathcal{D}}(\pi_M)\right] = \left(\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}\right) \cdot \frac{\sum_{\hat{\pi}\in\Pi^{\mathcal{D}}\setminus\{\pi_M,\pi'_{M'}\}} \frac{1}{\kappa(\hat{\pi})}}{\sum_{\hat{\pi}\in\Pi^{\mathcal{D}}} \frac{1}{\kappa(\hat{\pi})}} \geq 0.$$

The inequality is strict if both rules are active and at least one additional rule is active, which is equivalent to

$$\rho^{\mathcal{D}}(\pi'_{M'}) > 0 \qquad \text{and} \qquad \rho^{\mathcal{D}}(\pi_M) + \rho^{\mathcal{D}}(\pi'_{M'}) < 1.$$

If the two rules are outside the consideration set, their probabilities are both zero. Since the

solution to the strictly concave maximization problem in equation (1) is continuous in rule values, changes in the consideration set do not reverse the weak monotonicity.

Finally, consider changes in the costs that leave the consideration set unchanged. If the two rules are active, differentiating the preceding probability difference gives

$$\frac{\partial\left[\rho^{\mathcal{D}}(\pi'_{M'})-\rho^{\mathcal{D}}(\pi_M)\right]}{\partial\left(\frac{1}{\kappa(\pi'_{M'})}\right)}=\left(v^{\mathcal{D}}(\pi'_{M'})-\Lambda(\Pi^{\mathcal{D}})\right)\cdot\frac{\frac{2}{\kappa(\pi_M)}+\sum\limits_{\hat{\pi}\in\Pi^{\mathcal{D}}\setminus\{\pi_M,\pi'_{M'}\}}\frac{1}{\kappa(\hat{\pi})}}{\sum\limits_{\hat{\pi}\in\Pi^{\mathcal{D}}}\frac{1}{\kappa(\hat{\pi})}}>0,$$

and

$$\frac{\partial\left[\rho^{\mathcal{D}}(\pi'_{M'})-\rho^{\mathcal{D}}(\pi_M)\right]}{\partial\left(\frac{1}{\kappa(\pi_M)}\right)}=-\left(v^{\mathcal{D}}(\pi_M)-\Lambda(\Pi^{\mathcal{D}})\right)\cdot\frac{\frac{2}{\kappa(\pi'_{M'})}+\sum\limits_{\hat{\pi}\in\Pi^{\mathcal{D}}\setminus\{\pi_M,\pi'_{M'}\}}\frac{1}{\kappa(\hat{\pi})}}{\sum\limits_{\hat{\pi}\in\Pi^{\mathcal{D}}}\frac{1}{\kappa(\hat{\pi})}}<0.$$

If the two rules are inactive and remain outside the consideration set, both probabilities remain zero, so the corresponding monotonicities are weak. □

One could consider two decision problems with different value-maximizing rules that attain the same value, but different costs. In this setting, therefore, comparative statics are meaningful for comparing the probability of extracting different rules that differ only in their costs, while holding all other aspects of the decision problem constant. Because costs are rule-specific, ceteris paribus changes in costs while holding values fixed can be analyzed by introducing a bijection $\phi:\Pi(\mathcal{D})\to\Pi(\mathcal{D}')$.

Proposition 7. *Fix a decision problem $\mathcal{D}$ and two rules $\pi_M,\pi'_{M'}\in\Pi^{\mathcal{D}}$ such that $\kappa(\pi_M)>\kappa(\pi'_{M'})$ and $v^{\mathcal{D}}(\pi_M)\geq v^{\mathcal{D}}(\pi'_{M'})$. Consider a decision problem $\mathcal{D}'$ such that $\phi(\pi_M)=\pi'_{M'}$, $\phi(\pi'_{M'})=\pi_M$, and $\phi(\hat{\pi}_{\hat{M}})=\hat{\pi}_{\hat{M}}$ for every $\hat{\pi}_{\hat{M}}\in\Pi\setminus\{\pi_M,\pi'_{M'}\}$. Suppose that $v^{\mathcal{D}'}\left(\phi(\hat{\pi}_{\hat{M}})\right)=v^{\mathcal{D}}(\hat{\pi}_{\hat{M}})$ for every $\hat{\pi}_{\hat{M}}\in\Pi$, and that $\Pi^{\mathcal{D}}=\Pi^{\mathcal{D}'}$. Then:*

(i) $\rho^{\mathcal{D}'}\left(\phi(\pi_M)\right)>\rho^{\mathcal{D}}(\pi_M)$. Holding all other values and costs and the common consideration set fixed, this difference is weakly increasing in $v^{\mathcal{D}}(\pi_M)$, strictly so if $\Pi^{\mathcal{D}}\setminus\{\pi_M,\pi'_{M'}\}\neq\varnothing$, and is strictly increasing in $\frac{1}{\kappa(\pi'_{M'})}$ and strictly decreasing in $\frac{1}{\kappa(\pi_M)}$.

(ii) For every $\hat{\pi}_{\hat{M}}\in\Pi\setminus\{\pi_M\}$, $\rho^{\mathcal{D}'}\left(\phi(\hat{\pi}_{\hat{M}})\right)-\rho^{\mathcal{D}}(\hat{\pi}_{\hat{M}})\leq 0$, with strict inequality if $\rho^{\mathcal{D}}(\hat{\pi}_{\hat{M}})>0$ and $v^{\mathcal{D}}(\pi_M)>v^{\mathcal{D}}(\pi'_{M'})$. Holding all other values and costs and the common consideration set fixed, for every

$$\hat{\pi}_{\hat{M}}\in\Pi^{\mathcal{D}}\setminus\{\pi_M,\pi'_{M'}\},$$

the magnitude of the decrease is weakly increasing in $\frac{1}{\kappa(\hat{\pi}_{\hat{M}})}$ and $\frac{1}{\kappa(\pi'_{M'})}$, and weakly decreasing in $\frac{1}{\kappa(\pi_M)}$. All three monotonicities are strict if $v^{\mathcal{D}}(\pi_M)>v^{\mathcal{D}}(\pi'_{M'})$.

Proposition 7 compares two decision problems with the same consideration set whose value profiles are related by a bijection ϕ that exchanges π_M and $\pi'_{M'}$ while leaving every other rule unchanged. If $\kappa(\pi_M) > \kappa(\pi'_{M'})$ and $v^{\mathcal{D}}(\pi_M) \geq v^{\mathcal{D}}(\pi'_{M'})$, then assigning the value of π_M to the lower-cost rule $\phi(\pi_M) = \pi'_{M'}$ strictly increases its extraction probability: $\rho^{\mathcal{D}'}\big(\phi(\pi_M)\big) - \rho^{\mathcal{D}}(\pi_M) > 0$. Holding the remaining values, costs, and the common consideration set fixed, this increase is weakly larger when $v^{\mathcal{D}}(\pi_M)$ is higher, and strictly larger when the consideration set contains a rule other than π_M and $\pi'_{M'}$. The extraction probability of every other rule weakly decreases. For rules outside the exchanged pair, the magnitude of this decrease is weakly larger for lower-cost rules; the stated monotonicities are strict when $v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}}(\pi'_{M'})$.

We remark that Proposition 7 applies to $\pi_M = \pi^{\max}_{M^*}$, since $v^{\mathcal{D}}(\pi^{\max}_{M^*})$ attains the maximal value among all rules in any decision problem $\mathcal{D}$. In this case, when the highest-value rule attains a sufficiently low cost in $\mathcal{D}'$, some rules may be excluded from the consideration set, so that $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$ (see Lemma 3).

Proof. Because $\Pi^{\mathcal{D}} = \Pi^{\mathcal{D}'}$, swapping the values of the two rules changes the threshold by

$$\Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}}) = \frac{\left(\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}\right)\left(v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})\right)}{\kappa(\Pi^{\mathcal{D}})} \geq 0.$$

For the rule π_M, equation (4) gives

$$\begin{aligned}
&\rho^{\mathcal{D}'}\big(\phi(\pi_M)\big) - \rho^{\mathcal{D}}(\pi_M) \\
&= \frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})} \left[1 + \sum_{\hat{\pi}_{\hat{M}} \in \Pi^{\mathcal{D}} \setminus \{\pi_M, \pi'_{M'}\}} \frac{v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\hat{\pi}_{\hat{M}})}{\kappa(\hat{\pi}_{\hat{M}})}\right].
\end{aligned}$$

The term in square brackets can be written as

$$\begin{aligned}
&\kappa(\Pi^{\mathcal{D}})\left[v^{\mathcal{D}}(\pi'_{M'}) - \Lambda(\Pi^{\mathcal{D}'})\right] \\
&\quad + \left[\kappa(\Pi^{\mathcal{D}}) - \frac{1}{\kappa(\pi_M)}\right]\left[v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})\right],
\end{aligned}$$

which is strictly positive because $\phi(\pi'_{M'}) \in \Pi^{\mathcal{D}'}$. Since $\kappa(\pi_M) > \kappa(\pi'_{M'})$, the probability difference is strictly positive.

Holding the common consideration set fixed,

$$\begin{aligned}
&\frac{\partial}{\partial v^{\mathcal{D}}(\pi_M)}\left[\rho^{\mathcal{D}'}\big(\phi(\pi_M)\big) - \rho^{\mathcal{D}}(\pi_M)\right] \\
&= \frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})} \sum_{\hat{\pi}_{\hat{M}} \in \Pi^{\mathcal{D}} \setminus \{\pi_M, \pi'_{M'}\}} \frac{1}{\kappa(\hat{\pi}_{\hat{M}})} \geq 0.
\end{aligned}$$

The inequality is strict if the common consideration set contains a rule other than π_M and

$\pi'_{M'}$. Moreover,

$$\frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})}$$

is strictly increasing in $\frac{1}{\kappa(\pi'_{M'})}$ and strictly decreasing in $\frac{1}{\kappa(\pi_M)}$. This proves point (i).

For any $\hat{\pi}_{\hat{M}} \in \Pi^{\mathcal{D}} \setminus \{\pi_M, \pi'_{M'}\}$, we have

$$\begin{aligned}
&\rho^{\mathcal{D}'}\left(\phi(\hat{\pi}_{\hat{M}})\right) - \rho^{\mathcal{D}}(\hat{\pi}_{\hat{M}}) \\
&= -\frac{1}{\kappa(\hat{\pi}_{\hat{M}})} \frac{\left(\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}\right)\left(v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})\right)}{\kappa(\Pi^{\mathcal{D}})} \leq 0.
\end{aligned}$$

Rules outside the common consideration set have probability zero in both problems.

Finally,

$$\begin{aligned}
&\rho^{\mathcal{D}'}\left(\phi(\pi'_{M'})\right) - \rho^{\mathcal{D}}(\pi'_{M'}) \\
&= -\frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})} \left[1 + \sum_{\hat{\pi}_{\hat{M}} \in \Pi^{\mathcal{D}} \setminus \{\pi_M, \pi'_{M'}\}} \frac{v^{\mathcal{D}}(\pi'_{M'}) - v^{\mathcal{D}}(\hat{\pi}_{\hat{M}})}{\kappa(\hat{\pi}_{\hat{M}})}\right] < 0,
\end{aligned}$$

because the term in square brackets equals

$$\kappa(\Pi^{\mathcal{D}})\left[v^{\mathcal{D}}(\pi'_{M'}) - \Lambda(\Pi^{\mathcal{D}})\right] + \frac{v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})}{\kappa(\pi_M)} > 0.$$

Thus the probability of every rule other than π_M weakly decreases, with the stated strictness.

For every unchanged active rule, the magnitude of the decrease is

$$\frac{1}{\kappa(\hat{\pi}_{\hat{M}})}\left[v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})\right] \frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})}.$$

Holding all other inverse costs fixed,

$$\frac{\frac{1}{\kappa(\hat{\pi}_{\hat{M}})}}{\kappa(\Pi^{\mathcal{D}})}$$

is strictly increasing in $\frac{1}{\kappa(\hat{\pi}_{\hat{M}})}$, while

$$\frac{\frac{1}{\kappa(\pi'_{M'})} - \frac{1}{\kappa(\pi_M)}}{\kappa(\Pi^{\mathcal{D}})}$$

is strictly increasing in $\frac{1}{\kappa(\pi'_{M'})}$ and strictly decreasing in $\frac{1}{\kappa(\pi_M)}$. The corresponding monotonicities are therefore weak when $v^{\mathcal{D}}(\pi_M) = v^{\mathcal{D}}(\pi'_{M'})$ and strict when $v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}}(\pi'_{M'})$. This proves point (ii). □

OD(iii) Heterogeneity of Types

We now turn to an analysis of how the environment shapes disagreement. We begin by defining disagreement using a measure of behavioral heterogeneity proposed by Apesteguia and

Ballester (2026). Consider individuals, or types, indexed by $j \in \{1, \ldots, J\}$, each characterized by a stochastic choice function $\rho_j^{\mathcal{D}}(\cdot)$ of the form specified in equation (2). A population is represented by a probability distribution over types that assigns strictly positive mass to only a finite number of them, that is, an object of the form

$$\theta = \left[\theta_1, \theta_2, \ldots, \theta_J \,;\, \rho_1^{\mathcal{D}}, \rho_2^{\mathcal{D}}, \ldots, \rho_J^{\mathcal{D}}\right],$$

where θ_j denotes the population share of type $\rho_j^{\mathcal{D}}(\cdot)$ and $\sum_j \theta_j = 1$.[62] Disagreement is defined as the probability that, for a given decision problem $\mathcal{D}$, the rules extracted from two randomly sampled types differ,

$$CH^{\mathcal{D}}(\theta) = \sum_{j'} \theta_{j'} \sum_{j} \theta_j \sum_{\pi_M \in \Pi} \rho_{j'}^{\mathcal{D}}(\pi_M)\Big(1 - \rho_j^{\mathcal{D}}(\pi_M)\Big).$$

This measure can be decomposed to capture both within- and between-type variability, which is not only theoretically relevant but also provides a practical way to disentangle these two sources of heterogeneity in experimental settings:

$$CH^{\mathcal{D}}(\theta) = \sum_{j=1}^{J} \theta_j \left[\frac{|\Pi| - 1}{|\Pi|} - \sum_{\pi_M \in \Pi} \left(\rho_j^{\mathcal{D}}(\pi_M) - \frac{1}{|\Pi|}\right)^2\right] + \sum_{1 \le j < k \le J} \theta_j \theta_k \sum_{\pi_M \in \Pi} \left[\rho_j^{\mathcal{D}}(\pi_M) - \rho_k^{\mathcal{D}}(\pi_M)\right]^2.$$

The first term captures within-type heterogeneity, measuring how close each type in the population is to uniform choice, weighted by its prevalence. We denote this first component by $CH^{\mathcal{D}}_{\text{Within}}(\theta)$. The second term captures between-type heterogeneity, measuring the distance between each pair of distinct types in the population, again weighted by their prevalence. We denote this second component by $CH^{\mathcal{D}}_{\text{Between}}(\theta)$.

Consider a population $\theta = [\theta_1, \theta_2; \rho_1^{\mathcal{D}}, \rho_2^{\mathcal{D}}]$ in which types differ in their cost functions, with $\kappa_j(\cdot)$ denoting the cost function of type j, and let $\Pi_j^{\mathcal{D}}$ denote the consideration set of type j in decision problem $\mathcal{D}$. The following proposition provides sufficient conditions on the underlying parameters that govern disagreement, offering an ex ante characterization of when higher disagreement is to be expected. Importantly, while the rise in political polarization and disagreement has often been attributed to heterogeneity in information or preferences, Proposition 8 shows that disagreement need not rely on such heterogeneity: even over objective facts, disagreement can increase due to cognitive constraints and differences in the costs of noticing and encoding information. As rules become more costly, individuals may rely less on differences in value and possibly consider a broader set of rules, and consequently make more stochastic choices. Assume additionally that every inactive rule lies strictly below

[62] In Appendix D we discuss how to adapt the approach in Agranov et al. (2025) to identify different types.

type 2's consideration cutoff:

$$v^{\mathcal{D}}(\pi_M) < \Lambda_2^{\mathcal{D}} \qquad \text{for every } \pi_M \in \Pi \setminus \Pi_2^{\mathcal{D}},$$

where $\Lambda_2^{\mathcal{D}}$ denotes the cutoff associated with κ_2.

Proposition 8. *Fix a cost function $\kappa_2(\cdot)$ and a decision problem $\mathcal{D}$. Suppose that $\Pi_2^{\mathcal{D}}$ contains rules attaining at least two distinct values, that $v^{\mathcal{D}}(\pi_M) < \Lambda_2^{\mathcal{D}}$ for every $\pi_M \in \Pi \setminus \Pi_2^{\mathcal{D}}$, and that one of the following two conditions holds:*

(a) For every ordered pair $\pi_M, \pi'_{M'} \in \Pi_2^{\mathcal{D}}$ such that $v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}}(\pi'_{M'})$, we have $\rho_2^{\mathcal{D}}(\pi_M) > \rho_2^{\mathcal{D}}(\pi'_{M'})$.

(b) For every ordered pair $\pi_M, \pi'_{M'} \in \Pi_2^{\mathcal{D}}$ such that $v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}}(\pi'_{M'})$, we have $\rho_2^{\mathcal{D}}(\pi_M) < \rho_2^{\mathcal{D}}(\pi'_{M'})$.

Then there exists $\bar{\gamma} > 1$ such that, for every $\gamma \in (1, \bar{\gamma})$, if $\kappa_1(\cdot) = \gamma\kappa_2(\cdot)$, the following hold:

(i) $CH^{\mathcal{D}}_{\text{Between}}(\theta) > 0$. In case (a)*, $CH_1^{\mathcal{D}} > CH_2^{\mathcal{D}}$, whereas in case* (b)*, $CH_1^{\mathcal{D}} < CH_2^{\mathcal{D}}$.*

Moreover, there exists $\underline{\alpha} \in (0,1)$ such that, if $\mathcal{D}'$ differs from $\mathcal{D}$ only in that $v^{\mathcal{D}'}(\pi_M) = \alpha v^{\mathcal{D}}(\pi_M)$ for every $\pi_M \in \Pi$ and for $\alpha \in (\underline{\alpha}, 1)$. In case (a)*, $CH_j^{\mathcal{D}'} > CH_j^{\mathcal{D}}$, whereas in case* (b) *$CH_j^{\mathcal{D}'} < CH_j^{\mathcal{D}}$ for every j. In both cases, with population shares held fixed, between-type heterogeneity decreases:*

$$CH^{\mathcal{D}'}_{\text{Between}}(\theta) = \alpha^2 CH^{\mathcal{D}}_{\text{Between}}(\theta) < CH^{\mathcal{D}}_{\text{Between}}(\theta).$$

Point (i) in Proposition 8 links the emergence of disagreement to the cost function: higher costs make value differences less salient, shifting probability mass toward lower-value rules.[63] As the value of each rule becomes harder to discern and extraction probabilities are ordered so that higher-value rules are chosen more frequently by low-cost types, this generates more dispersed choices both within high-cost types and in the overall population, thereby increasing disagreement. By contrast, when higher-value rules are initially extracted less often, higher costs further concentrate choice on low value alternatives. In this case, although lower-value rules are extracted more frequently, disagreement is reduced.

Moreover, Proposition 8 shows that when the values of all rules are scaled down by a common factor $\alpha \in (0,1)$, a mechanism analogous to an increase in costs emerges. As utility differences shrink, lower-value rules perform relatively better and probability mass shifts

[63] We focus on linear transformations of the cost function, as rule-specific changes in costs yield results that depend on assumptions about their functional form and magnitude.

toward them. When low-value rules are already extracted more frequently due to a lower cost, reductions in differences in values lead to less disagreement, but at the cost of more mistakes. Conversely, when higher-value rules were initially extracted more frequently, this shift leads to greater disagreement and a dispersion of choices toward lower-value rules. This prediction is consistent with the empirical findings in Fréchette et al. (2024), who document that the likelihood that any two participants disagree on their predictions increases by 28 percentage points in high-noise relative to low-noise datasets.[64] Importantly, disagreement rises not only across participants but also within participants: the probability that two predictions by the same participant for the same x_M differ increases by 50 percent in high-noise relative to low-noise datasets.[65]

Proof. At $\gamma = 1$, the two types have the same cost function and hence

$$\Pi_1^{\mathcal{D}} = \Pi_2^{\mathcal{D}} \qquad \text{and} \qquad \rho_1^{\mathcal{D}} = \rho_2^{\mathcal{D}}.$$

Because Π is finite, the strict cutoff condition and the strict probability ranking are preserved under sufficiently small changes in γ. Continuity of the solution to equation (1) therefore implies that there exists $\bar{\gamma} > 1$ such that, for every $\gamma \in (1, \bar{\gamma})$, $\Pi_1^{\mathcal{D}} = \Pi_2^{\mathcal{D}}$, and the probability ranking in case (a) or case (b) holds for both types.

Fix such a γ. Proposition 4(ii) gives

$$\operatorname{sgn}\left(\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\right) = \operatorname{sgn}\left(v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\right)$$

for every $\pi_M \in \Pi_1^{\mathcal{D}}$. Thus the probability difference crosses zero once when rules are ordered by value. Moreover, Proposition 4(ii) implies that $\bar{v}^{\mathcal{D}}$ lies strictly between the smallest and largest values attained in the common consideration set, so the two probability distributions are distinct.

In case (a), probabilities are ordered in the same direction as values. The single-crossing property therefore implies that the sum of the largest components of $\rho_2^{\mathcal{D}}$ is weakly greater than the corresponding sum for $\rho_1^{\mathcal{D}}$ at every rank, with strict inequality at some rank. Hence $\rho_2^{\mathcal{D}}$ strictly majorizes $\rho_1^{\mathcal{D}}$. Since $x \mapsto x^2$ is strictly convex,

$$\sum_{\pi_M \in \Pi} \left(\rho_2^{\mathcal{D}}(\pi_M)\right)^2 > \sum_{\pi_M \in \Pi} \left(\rho_1^{\mathcal{D}}(\pi_M)\right)^2.$$

Consequently,

$$1 - \sum_{\pi_M \in \Pi} \left(\rho_1^{\mathcal{D}}(\pi_M)\right)^2 > 1 - \sum_{\pi_M \in \Pi} \left(\rho_2^{\mathcal{D}}(\pi_M)\right)^2.$$

In case (b), probabilities are ordered in the opposite direction from values. The same

[64] We note that, to further validate the results in the proposition, in low-noise datasets the highest-value rule is extracted with the highest frequency, in line with the premises of Proposition 8.

[65] A note of caution is due at this point, as the disagreement documented in that study concerns predictions for a single realization x_M, rather than disagreement over rules as formalized in our framework; a more direct comparison would require access to the underlying data, which are not yet available.

single-crossing argument therefore implies that $\rho_1^{\mathcal{D}}$ strictly majorizes $\rho_2^{\mathcal{D}}$. Hence

$$\sum_{\pi_M \in \Pi} \left(\rho_1^{\mathcal{D}}(\pi_M)\right)^2 > \sum_{\pi_M \in \Pi} \left(\rho_2^{\mathcal{D}}(\pi_M)\right)^2,$$

and thus

$$1 - \sum_{\pi_M \in \Pi} \left(\rho_1^{\mathcal{D}}(\pi_M)\right)^2 < 1 - \sum_{\pi_M \in \Pi} \left(\rho_2^{\mathcal{D}}(\pi_M)\right)^2.$$

Since the two distributions are distinct and $\theta_1, \theta_2 > 0$, the decomposition of disagreement gives

$$CH^{\mathcal{D}}_{\text{Between}}(\theta) = \theta_1 \theta_2 \sum_{\pi_M \in \Pi} \left[\rho_1^{\mathcal{D}}(\pi_M) - \rho_2^{\mathcal{D}}(\pi_M)\right]^2 > 0.$$

This proves point (i).

Finally, consider the uniform reduction in values. For each type j, dividing the objective under $\mathcal{D}'$ by α shows that

$$\arg\max_{\rho \in \Delta(\Pi)} \sum_{\pi_M \in \Pi} \left[\alpha v^{\mathcal{D}}(\pi_M)\rho(\pi_M) - \frac{\kappa_j(\pi_M)}{2}\rho(\pi_M)^2\right]$$

is equal to

$$\arg\max_{\rho \in \Delta(\Pi)} \sum_{\pi_M \in \Pi} \left[v^{\mathcal{D}}(\pi_M)\rho(\pi_M) - \frac{\kappa_j(\pi_M)}{2\alpha}\rho(\pi_M)^2\right].$$

Thus multiplying all values by $\alpha \in (0,1)$ is equivalent to multiplying all costs by $1/\alpha > 1$. Proposition 4(i) therefore implies $\Pi_j^{\mathcal{D}} \subseteq \Pi_j^{\mathcal{D}'}$, and hence $|\Pi_j^{\mathcal{D}'}| \geq |\Pi_j^{\mathcal{D}}|$.

The strict cutoff and probability-ranking conditions are preserved for α sufficiently close to one. Hence there exists $\underline{\alpha} \in (0,1)$ such that, for every $\alpha \in (\underline{\alpha}, 1)$, the consideration sets remain unchanged and the probability ranking is preserved. Applying the same single-crossing and majorization argument, in case (a) the distribution under $\mathcal{D}$ strictly majorizes that under $\mathcal{D}'$, whereas in case (b) the distribution under $\mathcal{D}'$ strictly majorizes that under $\mathcal{D}$. The stated inequalities for within-type heterogeneity follow from the strict convexity of $x \mapsto x^2$.

For the additional between-type comparisons (iii), the strict cutoff condition and continuity ensure that, for the local changes considered above,

$$\Pi_1^{\mathcal{D}} = \Pi_2^{\mathcal{D}} = \Pi_1^{\mathcal{D}'} = \Pi_2^{\mathcal{D}'}.$$

Equation (2) and $\kappa_1 = \gamma\kappa_2$ imply, for every $\pi_M \in \Pi_2^{\mathcal{D}}$,

$$\rho_1^{\mathcal{D}}(\pi_M) - \rho_2^{\mathcal{D}}(\pi_M) = (\gamma^{-1} - 1)\left[\rho_2^{\mathcal{D}}(\pi_M) - \frac{\kappa_2(\pi_M)^{-1}}{\sum_{\pi'_{M'} \in \Pi_2^{\mathcal{D}}} \kappa_2(\pi'_{M'})^{-1}}\right].$$

The bracketed terms are independent of γ, holding $\mathcal{D}$ and κ_2 fixed. They cannot all vanish: by equation (2), this would require all rules in $\Pi_2^{\mathcal{D}}$ to have the same value. Probabilities outside this set are zero. Consequently, with population shares fixed, $CH^{\mathcal{D}}_{\text{Between}}$ is a strictly positive constant times $(1 - \gamma^{-1})^2$, and is therefore strictly increasing in $\gamma > 1$ on the local interval.

Under $v^{\mathcal{D}'} = \alpha v^{\mathcal{D}}$, both each rule's value and the inverse-cost-weighted average value in equation (2) are multiplied by α. The base probability is unchanged and is the same for both types: normalizing inverse costs cancels the factor γ^{-1}. Consequently, for $j = 1, 2$ and every $\pi_M \in \Pi_2^{\mathcal{D}}$,

$$\begin{aligned}\rho_j^{\mathcal{D}'}(\pi_M) &= \frac{\kappa_2(\pi_M)^{-1}}{\sum_{\pi'_{M'} \in \Pi_2^{\mathcal{D}}} \kappa_2(\pi'_{M'})^{-1}} + \alpha \left[\rho_j^{\mathcal{D}}(\pi_M) - \frac{\kappa_2(\pi_M)^{-1}}{\sum_{\pi'_{M'} \in \Pi_2^{\mathcal{D}}} \kappa_2(\pi'_{M'})^{-1}} \right] \\ &= \alpha \rho_j^{\mathcal{D}}(\pi_M) + (1-\alpha) \frac{\kappa_2(\pi_M)^{-1}}{\sum_{\pi'_{M'} \in \Pi_2^{\mathcal{D}}} \kappa_2(\pi'_{M'})^{-1}}.\end{aligned}$$

Subtracting cancels the common second term and yields

$$\rho_1^{\mathcal{D}'}(\pi_M) - \rho_2^{\mathcal{D}'}(\pi_M) = \alpha \left[\rho_1^{\mathcal{D}}(\pi_M) - \rho_2^{\mathcal{D}}(\pi_M) \right].$$

This identity also holds outside $\Pi_2^{\mathcal{D}}$, where all four probabilities are zero. Hence, with population shares fixed,

$$\begin{aligned}CH_{\text{Between}}^{\mathcal{D}'}(\theta) &= \theta_1 \theta_2 \sum_{\pi_M \in \Pi} \left[\rho_1^{\mathcal{D}'}(\pi_M) - \rho_2^{\mathcal{D}'}(\pi_M) \right]^2 \\ &= \theta_1 \theta_2 \sum_{\pi_M \in \Pi} \left\{ \alpha \left[\rho_1^{\mathcal{D}}(\pi_M) - \rho_2^{\mathcal{D}}(\pi_M) \right] \right\}^2 \\ &= \alpha^2 \theta_1 \theta_2 \sum_{\pi_M \in \Pi} \left[\rho_1^{\mathcal{D}}(\pi_M) - \rho_2^{\mathcal{D}}(\pi_M) \right]^2 \\ &= \alpha^2 CH_{\text{Between}}^{\mathcal{D}}(\theta) < CH_{\text{Between}}^{\mathcal{D}}(\theta).\end{aligned}$$

The strict inequality follows from $\alpha \in (0, 1)$ and the positivity established above. Neither comparison requires distinguishing cases (a) and (b). □

OD(iv) Non-separability of second-highest-value rules: Either and Joint

We work in the two-light environment, $N = \{1, 2\}$ (red = 1, blue = 2), with $X_i = \{0, 1\}$, $X = \{0, 1\}^2$, $A = Y = \{0, 1\}$, payoffs $u(a, y) = \mathbf{1}[a = y]$, and $M^* = \{1, 2\}$. For each rule π_M and realization $x_M \in X_M$, define its conditional value by

$$v^{\mathcal{D}}(\pi_M \mid x_M) := \sum_{y \in Y} \sum_{a \in A} \pi_M(a \mid x_M) u(a, y) p(y \mid x_M).$$

The ex ante value defined in Section III(ii) can then be written as

$$v^{\mathcal{D}}(\pi_M) = \sum_{x_M \in X_M} v^{\mathcal{D}}(\pi_M \mid x_M) p(x_M).$$

We use the value-maximizing rule $\pi_{M^*}^{\max}$ defined in Section III(iii), which we take to be the unique value-maximizing rule of the decision problem, as in the experiment. We call π a *second-highest-value* rule if it maximizes $v^{\mathcal{D}}$ among rules with $v^{\mathcal{D}}(\pi) < v^{\mathcal{D}}(\pi_{M^*}^{\max})$, and we say

π is *one-dimensional* if it is measurable with respect to a single light. The DGP need not be deterministic, but we assume p has *full support*, i.e. $p(x, y) > 0$ for every $(x, y) \in X \times Y$.

Proposition 9. *Suppose that* Π *contains every deterministic rule, and that the optimal action is unique at every* $x \in X$*. Let the highest-value rule be* Either *or* Joint*, and suppose that* $p_X(x) > 0$ *for every* $x \in X$*. Then no one-dimensional deterministic rule attains the second-highest value: every such rule is strictly dominated by an admissible two-dimensional rule whose value remains strictly below* $v^{\mathcal{D}}(\pi^{\max}_{M^*})$*.*

Proof. Take Joint, so that $\pi^{\max}_{M^*}$ predicts 1 on $(0,0),(1,1)$ and 0 on $(0,1),(1,0)$ (Either only relabels these values). Suppose, for contradiction, that some one-dimensional rule π attains the second-highest value. Without loss it is measurable with respect to light 1, hence constant on each slice $\{x_1 = 0\} = \{(0,0),(0,1)\}$ and $\{x_1 = 1\} = \{(1,0),(1,1)\}$. Because $\pi^{\max}_{M^*}$ takes both values on each slice, a constant prediction must agree with it on one configuration of the slice and disagree on the other; thus π disagrees with $\pi^{\max}_{M^*}$ on exactly one configuration per slice, two in total. Now keep π unchanged except on one of these two "wrong" configurations, where we flip its prediction to match $\pi^{\max}_{M^*}$. The resulting rule is two-dimensional and agrees with $\pi^{\max}_{M^*}$ on one more configuration than π; since p has full support that configuration carries strictly positive mass, so the value rises strictly, while disagreement on the remaining configuration (also of positive mass) keeps the value strictly below $v^{\mathcal{D}}(\pi^{\max}_{M^*})$. This rule strictly dominates π while remaining below the highest attainable value, contradicting the assumption that π attains the second-highest value. □

The obstruction is a property of the highest-value rule and the population design, not of finite data: it contrasts with Or, and Inhibit, each constant on one slice of one light, for which a one-dimensional rule disagrees with the highest-value rule on a single configuration and can attain the second-highest value.

OE GENERALIZATION OF THE STOCHASTIC CHOICE FUNCTION

This appendix states the conditions under which the paper's comparative statics extend from the quadratic cost specification in equation (1) to a general convex cost function within the additive perturbed utility framework of Fudenberg et al. (2015).

$$\rho^{\mathcal{D}} = \operatorname*{argmax}_{\rho \in \Delta(\Pi)} \left\{ \sum_{\pi_M \in \Pi} \rho(\pi_M) v^{\mathcal{D}}(\pi_M) - \sum_{\pi_M \in \Pi} c(\rho(\pi_M), \kappa(\pi_M)) \right\}. \tag{17}$$

Assumption 1. *The set* Π *is finite, with* $|\Pi| \geq 2$*. For every* $\kappa > 0$*,* $c(\cdot, \kappa)$ *is finite, continuous, and strictly convex on* $[0,1]$*, with* $c(0,\kappa) = 0$*; it is differentiable on* $(0,1)$*, with*

$c_\rho(\cdot, \kappa)$ *continuous and strictly increasing; and*

$$\lim_{r \downarrow 0} c_\rho(r, \kappa) = 0.$$

For $z \in \mathbb{R}$, let

$$q(z, \kappa) := \operatorname*{argmax}_{r \in [0,1]} \{rz - c(r, \kappa)\}.$$

The maximizer is unique. Moreover, for every $\kappa > 0$, $q(\cdot, \kappa)$ is continuous and weakly increasing. It is strictly increasing over any interval on which it takes values in $(0, 1)$, and $q(z, \kappa) > 0$ if and only if $z > 0$.

For a value profile $w : \Pi \to \mathbb{R}$, define

$$Q_w(t) := \sum_{\pi_M \in \Pi} q(w(\pi_M) - t, \kappa(\pi_M))$$

and select the largest threshold satisfying the adding-up condition:

$$\Lambda(w) := \sup\{t \in \mathbb{R} : Q_w(t) \geq 1\}.$$

For a decision problem $\mathcal{D}$, we write $Q^{\mathcal{D}} := Q_{v^{\mathcal{D}}}$ and $\Lambda(\Pi^{\mathcal{D}}) := \Lambda(v^{\mathcal{D}})$, preserving the notation used in the main paper. Assumption 1 implies that problem (17) has a unique solution satisfying

$$\rho^{\mathcal{D}}(\pi_M) = q\Big(v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}}), \kappa(\pi_M)\Big), \qquad \sum_{\pi_M \in \Pi} \rho^{\mathcal{D}}(\pi_M) = 1. \tag{18}$$

Consequently,

$$\Pi^{\mathcal{D}} = \{\pi_M \in \Pi : v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}})\}. \tag{19}$$

Adding a common constant b to all rule values leaves extraction probabilities unchanged and raises the threshold by b.

To verify the representation, note that Q_w is continuous and weakly decreasing, tends to $|\Pi|$ as $t \to -\infty$, and equals zero when $t \geq \max_{\pi_M} w(\pi_M)$. Hence $Q_w(\Lambda(w)) = 1$. For each rule, $q(w(\pi_M) - \Lambda(w), \kappa(\pi_M))$ maximizes

$$r\Big(w(\pi_M) - \Lambda(w)\Big) - c(r, \kappa(\pi_M))$$

over $r \in [0, 1]$. Summing the inequalities from these scalar maximization problems over rules cancels the terms involving $\Lambda(w)$ because feasible distributions sum to one. The resulting distribution therefore solves problem (17); strict convexity of total cost gives uniqueness. Equation (19) and translation invariance follow immediately.

Robustness of the main-text results. Under Assumption 1:

(i) Propositions 1 and 2, as well as Proposition 3, continue to hold. Corollary 1 also continues to hold, except that as $\alpha \downarrow 0$ the extraction distribution converges to

$$\rho^0 = \arg\min_{\rho \in \Delta(\Pi)} \sum_{\pi_M \in \Pi} c(\rho(\pi_M), \kappa(\pi_M)) \,. \tag{20}$$

The minimizer is unique and has full support. Under $c(r, \kappa) = \kappa r^2/2$, equation (20) becomes

$$\rho^0(\pi_M) = \frac{1/\kappa(\pi_M)}{\sum_{\pi'_{M'} \in \Pi} 1/\kappa(\pi'_{M'})},$$

which is the low-stakes limit stated in Corollary 1.

(ii) Suppose additionally that

$$c(r, \gamma\kappa) = \gamma c(r, \kappa) \qquad \text{for all } r \in [0,1] \text{ and } \kappa, \gamma > 0, \tag{21}$$

and that $\kappa_1(\pi_M) = \gamma \kappa_2(\pi_M)$ for every π_M, with $\gamma > 1$. For type $j \in \{1, 2\}$, define

$$Q_j^{\mathcal{D}}(t) := \sum_{\pi_M \in \Pi} q\Big(v^{\mathcal{D}}(\pi_M) - t, \kappa_j(\pi_M)\Big) \,, \qquad \Lambda_j^{\mathcal{D}} := \sup\{t \in \mathbb{R} : Q_j^{\mathcal{D}}(t) \geq 1\}.$$

If the high-cost consideration set $\Pi_1^{\mathcal{D}}$ contains at least two distinct value levels, Proposition 4 and Corollary 2 continue to hold. In particular, the crossing value is

$$\bar{v}^{\mathcal{D}} = \frac{\gamma \Lambda_2^{\mathcal{D}} - \Lambda_1^{\mathcal{D}}}{\gamma - 1}. \tag{22}$$

OE(i) PROOFS OF THE MAIN-TEXT RESULTS UNDER GENERAL COSTS

Proof of Proposition 1 under general costs. The consideration set is nonempty. If $\widehat{\pi}$ is active, then

$$v^{\mathcal{D}}(\widehat{\pi}) > \Lambda(\Pi^{\mathcal{D}}).$$

Maximality of $\pi_{M^*}^{\max}$ therefore implies $v^{\mathcal{D}}(\pi_{M^*}^{\max}) > \Lambda(\Pi^{\mathcal{D}})$, proving that the highest-value rule receives positive probability.

For the comparative static, let

$$b := v^{\mathcal{D}'}(\pi_{M^*}^{\max}) - v^{\mathcal{D}}(\pi_{M^*}^{\max})$$

and define $\widehat{v}(\pi_M) := v^{\mathcal{D}'}(\pi_M) - b$. Translation invariance implies that $\widehat{v}$ and $v^{\mathcal{D}'}$ induce the same extraction probabilities. The value-gap condition in Proposition 1 is equivalent to

$$\widehat{v}(\pi_{M^*}^{\max}) = v^{\mathcal{D}}(\pi_{M^*}^{\max}), \qquad \widehat{v}(\pi_M) < v^{\mathcal{D}}(\pi_M) \quad \text{for every } \pi_M \neq \pi_{M^*}^{\max}.$$

Consequently, monotonicity of q gives

$$Q_{\widehat{v}}(t) \le Q^{\mathcal{D}}(t) \qquad \text{for every } t,$$

and hence $\Lambda(\widehat{v}) \le \Lambda(\Pi^{\mathcal{D}})$. Using the threshold representation for the highest-value rule then yields

$$\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) = q\Big(v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \Lambda(\widehat{v}), \kappa(\pi^{\max}_{M^*})\Big) \ge \rho^{\mathcal{D}}(\pi^{\max}_{M^*}).$$

If $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$, some competing rule is active and therefore has an interior probability. That rule loses value strictly under $\widehat{v}$, so

$$Q_{\widehat{v}}\Big(\Lambda(\Pi^{\mathcal{D}})\Big) < 1.$$

It follows that $\Lambda(\widehat{v}) < \Lambda(\Pi^{\mathcal{D}})$. Because the highest-value rule's previous extraction probability is interior, strict monotonicity of q gives the strict probability comparison. □

Proof of Corollary 1 under general costs. Utility scaling implies that every rule value is multiplied by α. For the fixed value profile $v^{\mathcal{D}}$, let Λ_α and Π_α denote the threshold and consideration set under the profile $\alpha v^{\mathcal{D}}$, and define

$$\tau_\alpha := \frac{\Lambda_\alpha}{\alpha}.$$

The function

$$H_\alpha(t) := \sum_{\pi_M \in \Pi} q\Big(\alpha\Big(v^{\mathcal{D}}(\pi_M) - t\Big), \kappa(\pi_M)\Big)$$

has τ_α as its largest level-one root. If $\alpha_2 > \alpha_1$, then at $t = \tau_{\alpha_1}$ every positive argument is weakly enlarged, while every nonpositive argument continues to generate zero probability. Thus $H_{\alpha_2}(\tau_{\alpha_1}) \ge 1$, which implies $\tau_{\alpha_2} \ge \tau_{\alpha_1}$. Since

$$\pi_M \in \Pi_\alpha \quad \Longleftrightarrow \quad v^{\mathcal{D}}(\pi_M) > \tau_\alpha,$$

the consideration set contracts as incentives rise and expands as they fall.

The probability comparisons now follow from Proposition 1. For $\alpha < 1$, if the highest-value rule's probability at $\alpha = 1$ is below one, the support expansion just established ensures that its probability under the compressed profile is also below one. Applying Proposition 1 in the reverse direction therefore gives the claimed strict inequality.

Let $V := \max_{\pi_M} v^{\mathcal{D}}(\pi_M)$, and let ρ_α denote the solution under the profile $\alpha v^{\mathcal{D}}$. Total cost is bounded on the compact simplex. Comparing the optimum with a distribution supported on the value-maximizing rules gives

$$V - \sum_{\pi_M \in \Pi} \rho_\alpha(\pi_M) v^{\mathcal{D}}(\pi_M) = O(1/\alpha).$$

Because the value-maximizing rule is unique, all mass outside $\pi^{\max}_{M^*}$ vanishes as $\alpha \to \infty$.

As $\alpha \downarrow 0$, the objective converges uniformly on the simplex to the negative of total cost. Strict convexity therefore implies that ρ_α converges to the unique minimizer ρ^0 in equation (20). This minimizer has full support. Indeed, suppose that $\rho^0(\pi_M) = 0$ while

$\rho^0(\pi'_{M'}) = r > 0$. The right marginal cost at zero is zero, whereas strict convexity implies that the left marginal cost $c'_-(r, \kappa(\pi'_{M'}))$ is strictly positive. Moving a small amount of probability from $\pi'_{M'}$ to π_M therefore has directional derivative

$$-c'_-(r, \kappa(\pi'_{M'})) < 0,$$

contradicting cost minimization. Under the quadratic specification, the first-order conditions $\kappa(\pi_M)\rho^0(\pi_M) = \mu$ give exactly the low-stakes limit in Corollary 1. □

Proof of Proposition 2 under general costs. At the old threshold, the random-rule summand is unchanged and every other summand weakly falls. Since $\rho^{\mathcal{D}}(\pi^R) < 1$, at least one non-random rule is active. Its probability is interior and its value falls strictly, so

$$Q^{\mathcal{D}'}\Big(\Lambda(\Pi^{\mathcal{D}})\Big) < 1.$$

Therefore $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$. The random rule's old probability is interior, so strict monotonicity of q at its unchanged value gives $\rho^{\mathcal{D}'}(\pi^R) > \rho^{\mathcal{D}}(\pi^R)$. □

Proof of Proposition 3 under general costs. Evaluate the enlarged-menu adding-up function at the old threshold. The old terms sum to one and every new term is nonnegative. Hence

$$\Lambda(\Pi^{\mathcal{D}'}) \geq \Lambda(\Pi^{\mathcal{D}}),$$

and monotonicity of q implies that every old active rule receives weakly less probability.

If a new rule is active at the new threshold, equality of thresholds would make the old terms sum to one while a new term is strictly positive. Continuity would then imply that the enlarged-menu adding-up function remains above one at some threshold above the old one, contradicting the largest-root convention. The threshold therefore rises strictly. Every old active rule then loses probability strictly. When the old solution is not pure, this follows from strict monotonicity of q at each old active probability. When it is pure, the largest-root convention places the old threshold at the upper endpoint of the interval on which that rule receives probability one, so a strict threshold increase again lowers its probability. □

Proof of Proposition 4 under general costs. Under equation (21) and $\kappa_1(\pi_M) = \gamma\kappa_2(\pi_M)$, the two types' probabilities can be written using the low-cost type's inverse marginal cost:

$$\rho_1^{\mathcal{D}}(\pi_M) = q\left(\frac{v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}}{\gamma}, \kappa_2(\pi_M)\right), \qquad \rho_2^{\mathcal{D}}(\pi_M) = q\Big(v^{\mathcal{D}}(\pi_M) - \Lambda_2^{\mathcal{D}}, \kappa_2(\pi_M)\Big).$$

The high-cost consideration set contains at least two rules, so every high-cost active probability is interior. At $\Lambda_1^{\mathcal{D}}$, replacing each positive argument $(v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}})/\gamma$ by $v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}$ strictly raises every high-cost active summand. The low-cost adding-up function is therefore strictly above one at $\Lambda_1^{\mathcal{D}}$, which gives

$$\Lambda_1^{\mathcal{D}} < \Lambda_2^{\mathcal{D}}.$$

The support characterization immediately implies $\Pi_2^{\mathcal{D}} \subseteq \Pi_1^{\mathcal{D}}$.

For every $\pi_M \in \Pi_1^{\mathcal{D}}$, strict monotonicity of q at the high-cost type's interior probability gives

$$\begin{aligned}\operatorname{sgn}\Big(\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\Big) &= \operatorname{sgn}\left[v^{\mathcal{D}}(\pi_M) - \Lambda_2^{\mathcal{D}} - \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}}{\gamma}\right] \\ &= \operatorname{sgn}\Big(v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\Big),\end{aligned}$$

where $\bar{v}^{\mathcal{D}}$ is defined in equation (22). A rule outside $\Pi_1^{\mathcal{D}}$ has value weakly below $\Lambda_1^{\mathcal{D}} < \Lambda_2^{\mathcal{D}}$, so both types assign it probability zero.

The probability differences sum to zero. Because the high-cost support contains at least two distinct value levels, those differences cannot all be zero. Their single-crossing sign therefore requires at least one supported value below $\bar{v}^{\mathcal{D}}$ and at least one above it. Since a value-maximizing rule is always active, this establishes

$$\min_{\pi_M \in \Pi_1^{\mathcal{D}}} v^{\mathcal{D}}(\pi_M) < \bar{v}^{\mathcal{D}} < v^{\mathcal{D}}(\pi_{M^*}^{\max}).$$

It remains to establish first-order stochastic dominance. Put $d(\pi_M) := \rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)$. If $z < \bar{v}^{\mathcal{D}}$ and $F_1^{\mathcal{D}}(z) > 0$, every term in

$$\sum_{v^{\mathcal{D}}(\pi_M) \le z} d(\pi_M)$$

is nonpositive and at least one is strictly negative. Hence $F_2^{\mathcal{D}}(z) < F_1^{\mathcal{D}}(z)$. If $z \ge \bar{v}^{\mathcal{D}}$ and $F_1^{\mathcal{D}}(z) < 1$, some high-cost active rule has value above z, and every corresponding tail difference is strictly positive. Since total probability differences sum to zero,

$$F_2^{\mathcal{D}}(z) - F_1^{\mathcal{D}}(z) = -\sum_{v^{\mathcal{D}}(\pi_M) > z} d(\pi_M) < 0.$$

This is precisely $\rho_2^{\mathcal{D}} \succ_{\mathrm{FOSD}}^{v^{\mathcal{D}}} \rho_1^{\mathcal{D}}$. □

Proof of Corollary 2 under general costs. A unique value-maximizing rule has value strictly above $\bar{v}^{\mathcal{D}}$, so the sign characterization just proved gives

$$\rho_2^{\mathcal{D}}(\pi_{M^*}^{\max}) > \rho_1^{\mathcal{D}}(\pi_{M^*}^{\max}).$$

If $v^{\mathcal{D}}(\pi^R) < \bar{v}^{\mathcal{D}}$ and the random rule is active for type 1, the same sign characterization gives a strict decline under type 2. If it is inactive for type 1, support inclusion makes it inactive for both types, which gives the weak comparison. □

OE(ii) ADDITIONAL ASSUMPTIONS FOR THE APPENDIX RESULTS

The additional appendix results compare rules with different extraction costs and characterize the interaction between value and cost. In addition to Assumption 1, we maintain the following conditions.

Assumption 2 (Additional cost conditions)**.** *For every relevant pair $\kappa_L < \kappa_H$:*

(i) *Marginal extraction costs are ordered:*

$$c_\rho(r, \kappa_L) \leq c_\rho(r, \kappa_H) \qquad \textit{for every } r \in (0, 1).$$

Whenever a strict comparison is stated, the ordering is strict at the relevant probability.

(ii) *Whenever local comparative statics are considered, c is twice continuously differentiable in its first argument, $c_{\rho\rho}(r, \kappa) > 0$, and c_ρ is continuously differentiable in κ when costs vary.*

(iii) *Define the probability advantage of the lower-cost rule at net value z by*

$$D_{LH}(z) := q(z, \kappa_L) - q(z, \kappa_H).$$

The function D_{LH} is weakly increasing over the relevant net-value range. At points at which both probabilities are interior, this is equivalent to

$$\frac{1}{c_{\rho\rho}(q(z, \kappa_L), \kappa_L)} \geq \frac{1}{c_{\rho\rho}(q(z, \kappa_H), \kappa_H)}.$$

If a truncation at zero or one binds, monotonicity of D_{LH} is imposed directly on the relevant range.

(iv) *When types satisfy*

$$\kappa_1(\pi_M) = \gamma \kappa_2(\pi_M) \qquad \textit{for every } \pi_M \in \Pi, \qquad \gamma > 1,$$

costs are homogeneous of degree one:

$$c(r, \gamma\kappa) = \gamma c(r, \kappa) \qquad \textit{for every } r \in [0, 1] \textit{ and } \kappa, \gamma > 0.$$

Local comparisons are understood on regions in which the relevant rules are active and the consideration set is locally constant. The marginal-cost ordering gives κ its intended interpretation as an index of extraction difficulty. The monotonicity of D_{LH} is the substantive value–cost complementarity restriction: as a rule's net value increases, the probability advantage associated with its lower extraction cost does not decline.

OF INVARIANCE AND CAUSAL INTERPRETATION

The factorization defining $\mathcal{P}_{m^*}$ imposes two distinct restrictions. First, m^* is stable as p_X varies across datasets. Second, X_{M^*} is sufficient for predicting Y: conditional on X_{M^*}, the remaining observables contain no additional information about the outcome. Neither

restriction is, by itself, a statement of causality. This section relates the maintained statistical structure to a causal representation and clarifies when an invariant model may omit observed causal parents. Let

$$\mathcal{P}^{+}_{m^*} := \{p \in \mathcal{P}_{m^*} : p_X(x) > 0 \text{ for every } x \in X\}.$$

Proposition 10 (Invariance and omitted causal parents)**.** *The following statements hold.*

(i) A model M is invariant over $\mathcal{P}^{+}_{m^*}$ if and only if $M^* \subseteq M$. Hence M^* is the unique inclusion-minimal invariant model over $\mathcal{P}^{+}_{m^*}$.

(ii) Let $P = \mathrm{PA}(Y) \subseteq N$, suppose $M^* \subseteq P$, and set $R = P \setminus M^*$. If both $p(Y \mid X_P)$ and $p(X_R \mid X_{M^*})$ are stable across datasets, then M^* is invariant. Thus, M^* may omit observable causal parents whenever their conditional distribution given X_{M^*} is stable across datasets.

Proof. If $M^* \subseteq M$, the defining factorization of $\mathcal{P}_{m^*}$ gives $p(y \mid x_M) = m^*(y \mid x_{M^*})$, independently of p_X. Conversely, let $k \in M^* \setminus M$. By the inclusion-minimality of M^*, there exist two realizations of X_{M^*} that agree outside coordinate k but induce different distributions under m^*. Because p_X ranges over all full-support distributions in $\mathcal{P}^{+}_{m^*}$, two such distributions can place arbitrarily different conditional mass on these realizations given the same x_M. The resulting distributions of $Y \mid X_M = x_M$ differ, so M is not invariant. This proves (i).

For (ii), iterated conditioning gives

$$p(y \mid x_{M^*}) = \sum_{x_R \in X_R} p(y \mid x_{M^*}, x_R) p(x_R \mid x_{M^*}).$$

Because $(X_{M^*}, X_R) = X_P$, both terms in the mixture are invariant by assumption. □

Part (i) exploits the richness of $\mathcal{P}^{+}_{m^*}$: arbitrary variation in p_X rules out spurious invariance. Part (ii) shows why an omitted cause may remain hidden over a narrower set of environments. Identifying the full parent set therefore requires variation that shifts each omitted parent's conditional distribution and changes the induced distribution of Y (Peters et al., 2016). When $R \neq \varnothing$, part (ii) concerns the invariant reduced conditional $p(Y \mid X_{M^*})$. The omitted parents may still contain predictive information within an environment, so this case need not satisfy the stronger factorization defining $\mathcal{P}_{m^*}$. Allowing it therefore requires interpreting m^* as an invariant reduced-form DGP across those datasets, rather than as a conditionally sufficient representation of the full joint distribution. Under the maintained definition of $\mathcal{P}_{m^*}$, by contrast, M^* coincides with the unique minimal invariant predictive model.

OG Experimental Instructions

This appendix reproduces the participant-facing instructions used in the two-light and three-light versions of the experiment. The wording was the same across treatments within each version. The two versions differed only where indicated below. Graphical examples, response fields, and navigation controls from the online interface are omitted.

OG(i) Introduction

Welcome to the experiment!

Thank you for participating in this experiment on individual decision-making. In this study, we are interested in the choices you make and the processes behind your decisions.

This experiment takes around **30 minutes** and consists of **5 tasks**, each divided into two parts.

- In **Part 1**, you will receive information.
- In **Part 2**, you will make predictions about the information you received in Part 1.

What you **earn** depends on your decisions, and on chance. Your **baseline payment is £3.00** in the two-light version and **£3.20** in the three-light version. On top of that, one decision is randomly selected and if your answer is correct you will receive an **additional £1.50**. The **maximum total payment** is therefore **£4.50** in the two-light version and **£4.70** in the three-light version.

Please complete the tasks using only your own judgment

This is a research study in which we are genuinely interested in your responses.

Please do not rely on AI tools. These tools are unlikely to be useful in this experiment and may harm the scientific validity of the study.

OG(ii) Task Instructions

The instructions were presented on five consecutive slides. Participants could move backward and forward between slides before continuing.

Slide 1: Part 1 of each task

- In this experiment, you will complete **5 tasks**, each generated by a machine. In **Part 1** of each task, you will learn about observations **generated by the machine**. In **Part 2**, you will make predictions.
- **Each machine generates 27 observations. Each observation is independent of the other rows.** For each observation:
  - You can see if the red light is on or off.
  - You can see if the blue light is on or off.
  - In the three-light version, you can also see if the green light is on or off.
  - You cannot observe other lights, which could be on or off.
  - The machine either makes a sound or does not make a sound.

Slide 2: Prediction rules

- The lights and sounds may or may not be related. Even if a pattern exists, **the same observable light configuration may not always produce the same sound**, as unobservable lights may also affect the sound.
- Even if a pattern exists, **not all the observable lights are necessarily informative about the sound**.

In the example shown to participants, there was a relationship based only on the blue light, with the sound occurring four out of five times when the blue light was on.

Slide 3: Your objective

- **Your task in Part 1** is to learn the relationship (if one exists) between the observable lights and the sound to make predictions about the sound in Part 2.
- You have a maximum of **3 minutes** to observe data and take notes before making predictions. Keep in mind that the time only includes the learning phase, not the predictions that you will make in Part 2.
- In Part 2, you will not have access to the data, but you will be able to access your notes. Notes have a **maximum length of 75 characters**. Only predictions count for payoffs: not your notes or the time you take to complete the task.

- **Your task in Part 2** is to use your notes to make predictions about the sound.

Participants then practiced writing a note about the relationship involving the blue light. They were told: "This example is for your understanding only and does not count for payoffs."

Slide 4: Prediction interface

- In **Part 2** you will face **12 prediction trials** in the two-light version and **16 prediction trials** in the three-light version, with varying light configurations. In each trial, **your goal is to predict the corresponding sound**.
- You are given an example of one prediction trial related to the previous example. Remember that this example is for your understanding only and does not count for payoffs.

Slide 5: Payment

- You will face **5 tasks**, each generated by a machine. **For payment, one machine and one of your predictions for that machine will be randomly selected.**
- If your prediction is correct, **£1.50** will be added to your earnings.
- The payment example is for your understanding only and does not count for payoffs.

OG(iii) Comprehension Check

Participants were required to answer all five questions correctly before beginning the experiment. Incorrect answers were highlighted and the questions were shown again.

1. **Which of the following describes the setup of the experiment?**
    a. There are 5 tasks. All observations in the same task are generated by the same machine.
    b. There are 5 tasks. Different observations in the same task can be generated by different machines.
    c. The machine switches when predictions are correct.
    d. The machine changes whenever the lights change.
2. **In a given task, for the same light configuration (e.g., blue on and red on), will the sound always be the same?**

a. Yes, the sound is always identical when the lights match.

b. No, the sound may vary even with the same observable lights, due to the presence of unobservable lights.

c. The sound depends on previous predictions.

d. The rule changes after some trials.

3. **Do the lights in one row affect the sound in another observation?**

a. Yes, they do.

b. Yes, the machine learns over time.

c. No, each observation is independent.

d. Only in a given task, but not across tasks.

4. **What is your objective?**

a. Identify which machine is operating.

b. Predict whether the machine will make a sound.

c. Change the lights to influence the outcome.

d. Guess the overall pattern of the lights in the machine.

5. **How will your additional payment for the prediction stage be determined?**

a. You will be paid only for your last prediction.

b. A random observation from each machine will be selected and you will be paid if your prediction is correct.

c. A random machine will be selected, a random observation from that machine will be drawn, and you will be paid if your prediction is correct.

d. No additional payment, you are only paid a participation fee.